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Kim et al.

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(54) **MEMORY ARRAYS, AND METHODS OF FORMING MEMORY ARRAYS**

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See application file for complete search history.

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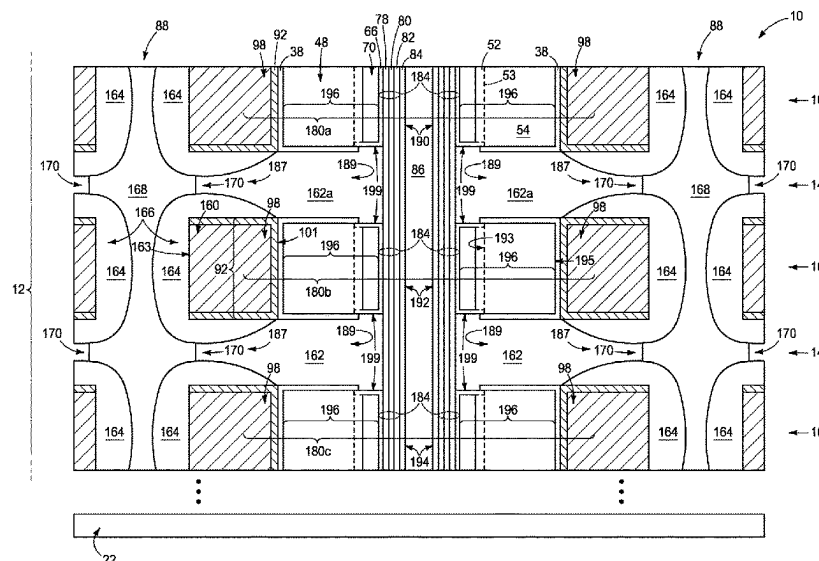
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(57) **ABSTRACT**

Some embodiments include a method of forming an assembly (e.g., a memory array). A first opening is formed through a stack of alternating first and second levels. The first levels contain silicon nitride, and the second levels contain silicon dioxide. Some of the silicon dioxide of the second levels is replaced with memory cell structures. The memory cell structures include charge-storage regions adjacent charge-blocking regions. Tunneling material is formed within the first opening, and channel material is formed adjacent the tunneling material. A second opening is formed through the stack. The second opening extends through remaining portions of the silicon dioxide, and through the silicon nitride. The remaining portions of the silicon dioxide are removed to form cavities. Conductive regions are formed within the cavities. The silicon nitride is removed to form voids between the conductive regions. Some embodiments include memory arrays.

8 Claims, 28 Drawing Sheets



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division of application No. 15/948,639, filed on Apr. 9, 2018, now Pat. No. 10,497,715.

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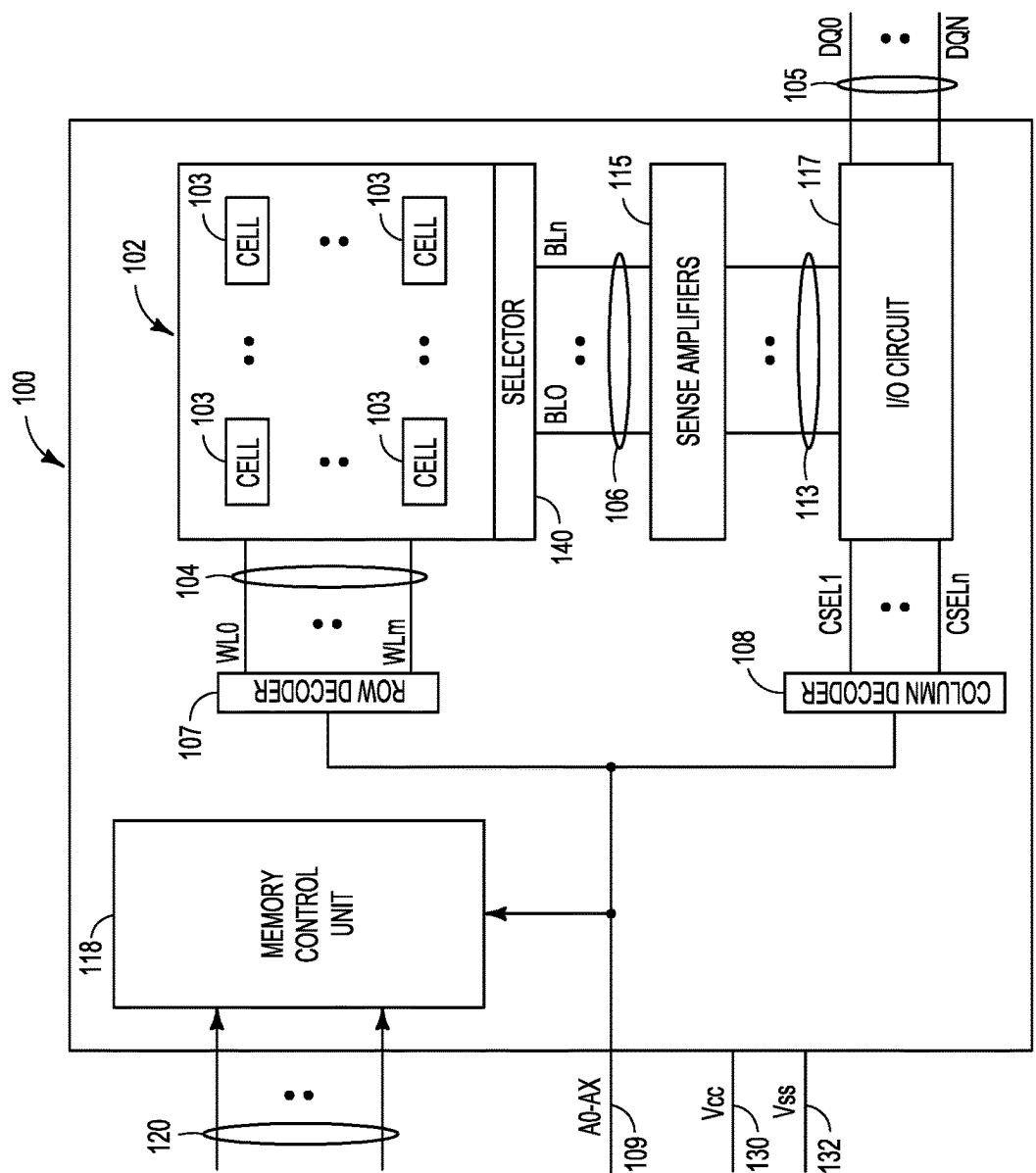
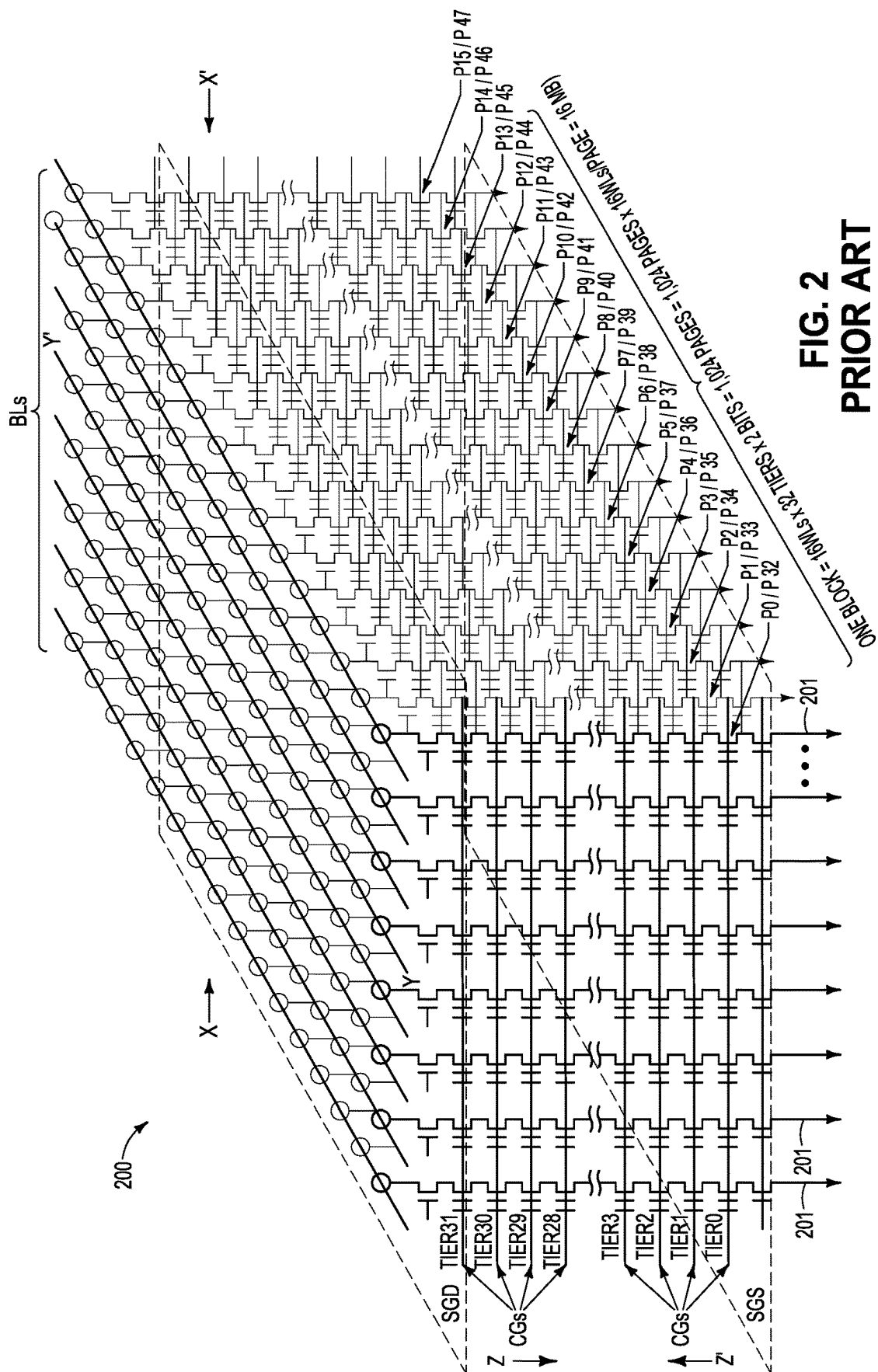
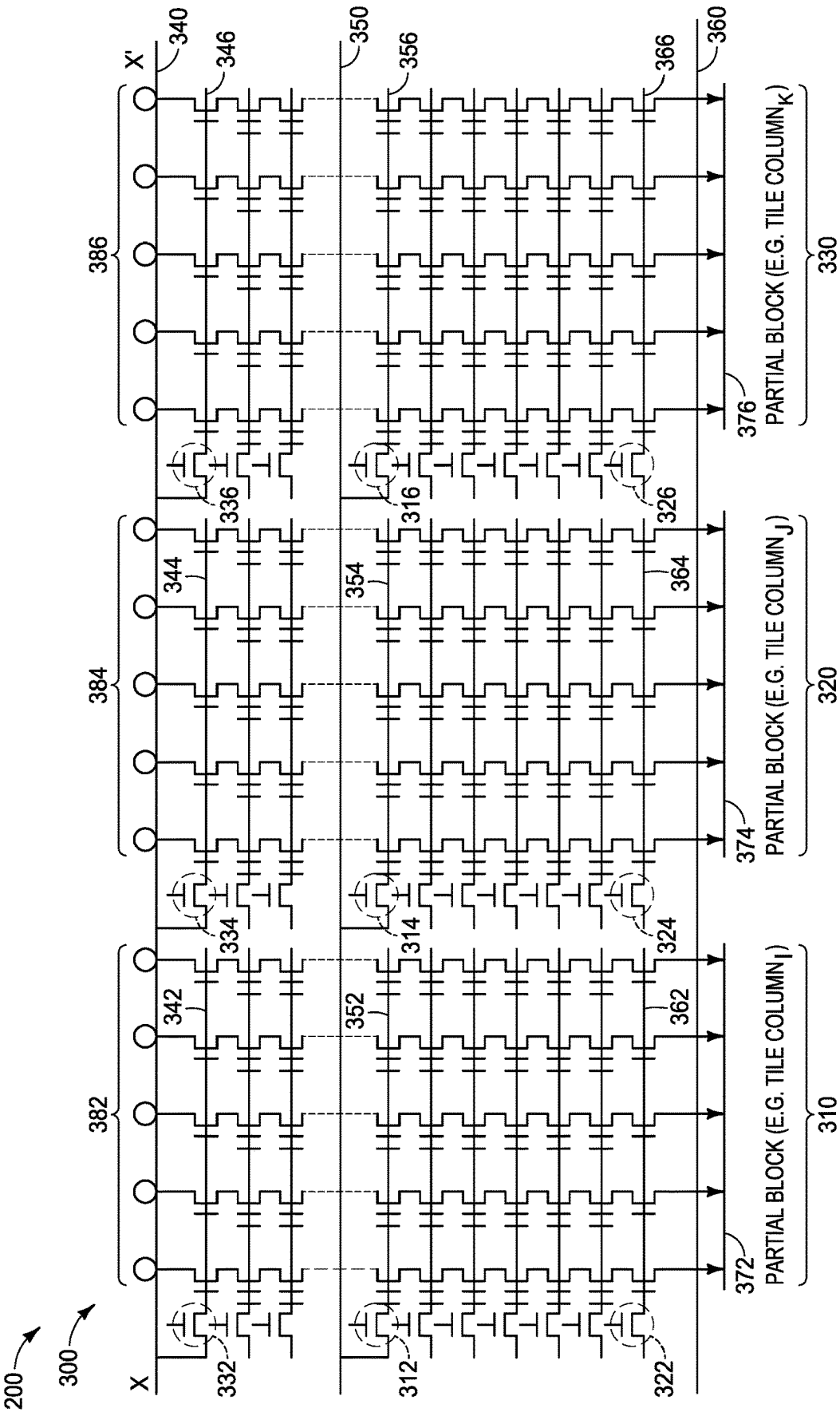


FIG. 1
PRIOR ART





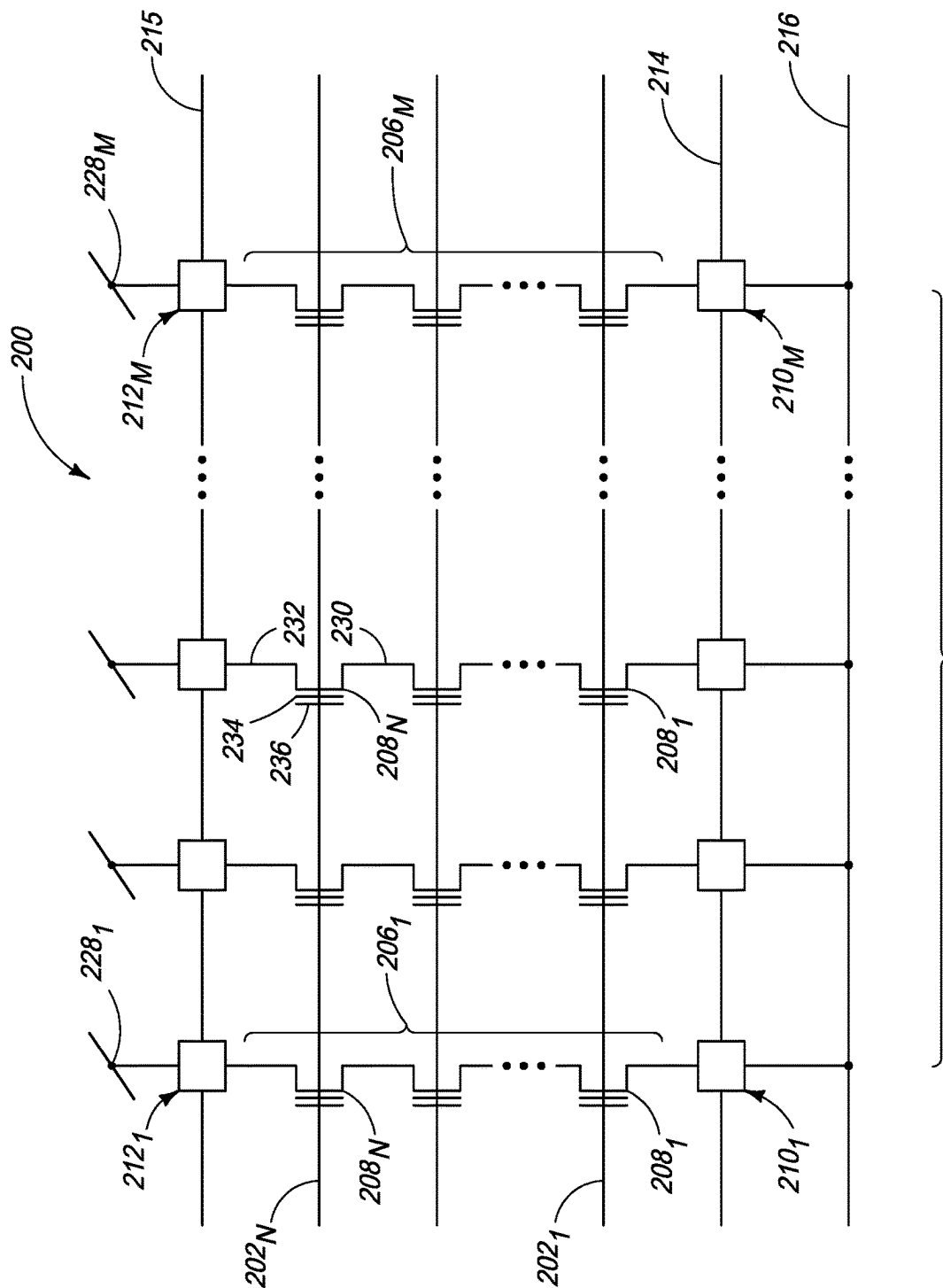
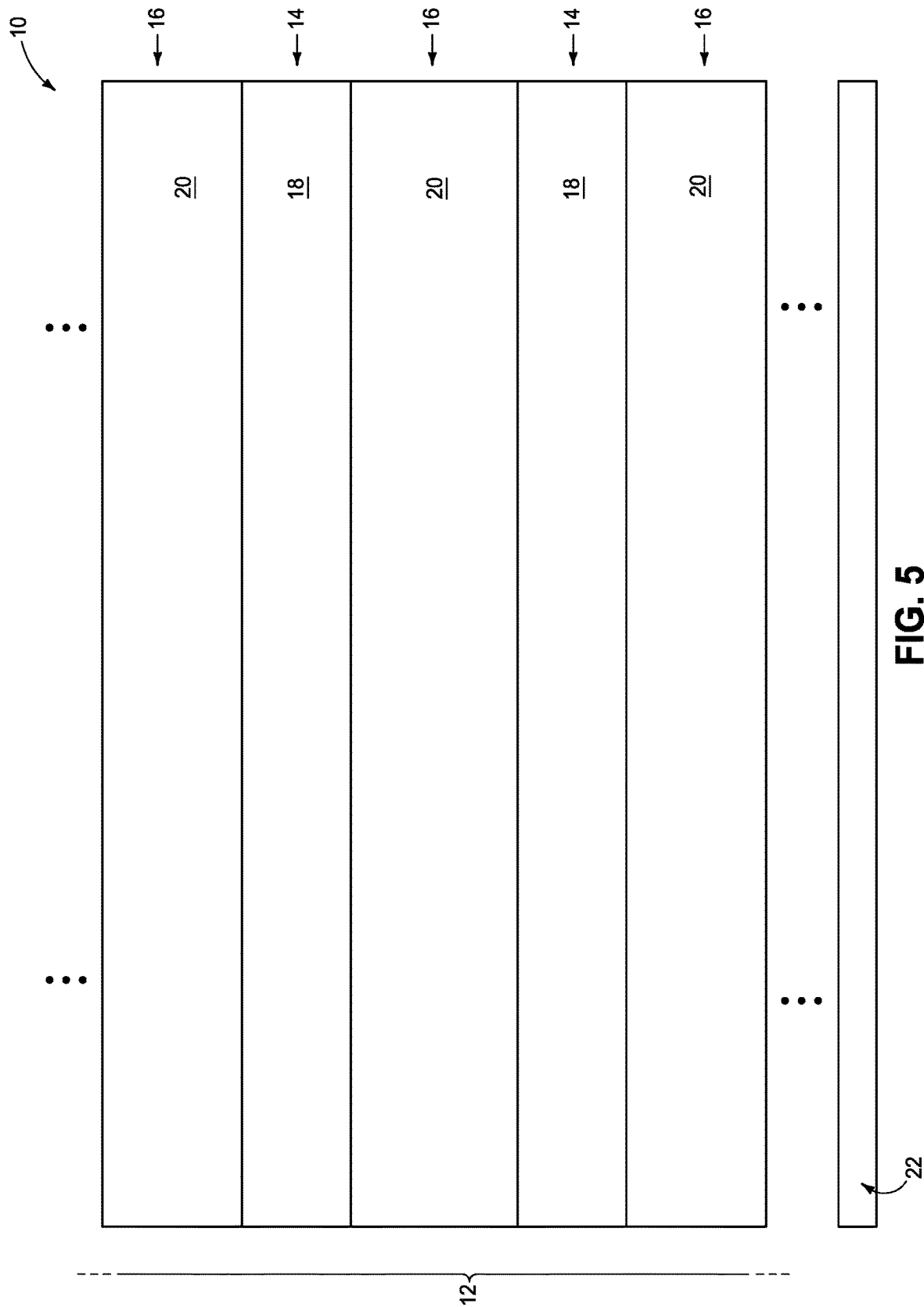
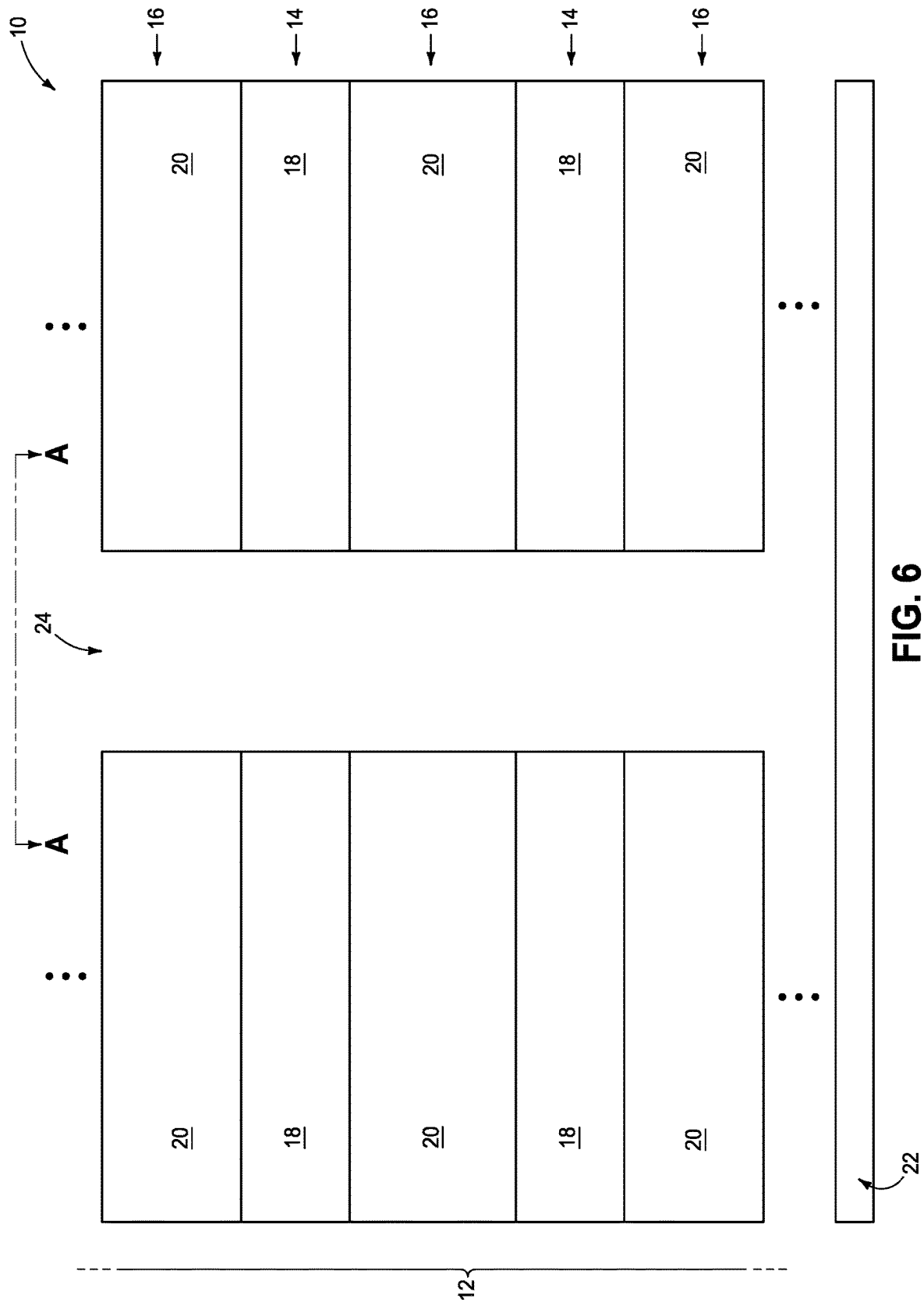


FIG. 4
PRIOR ART





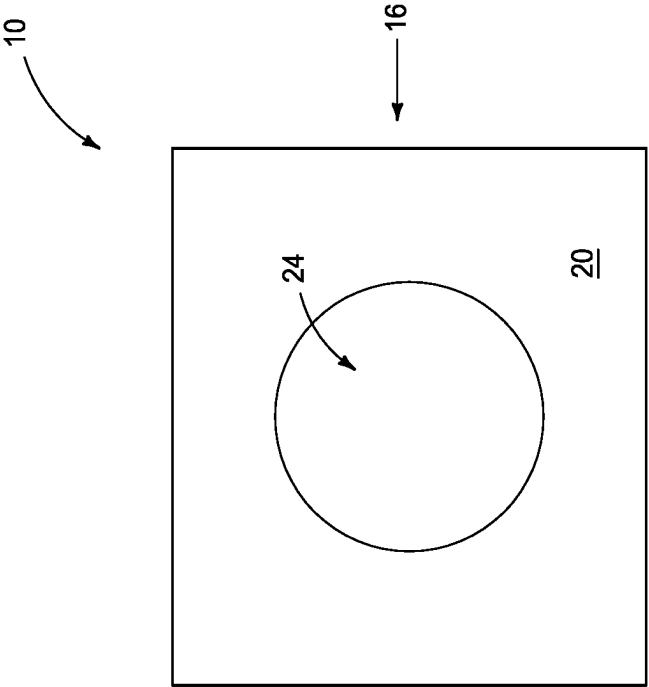
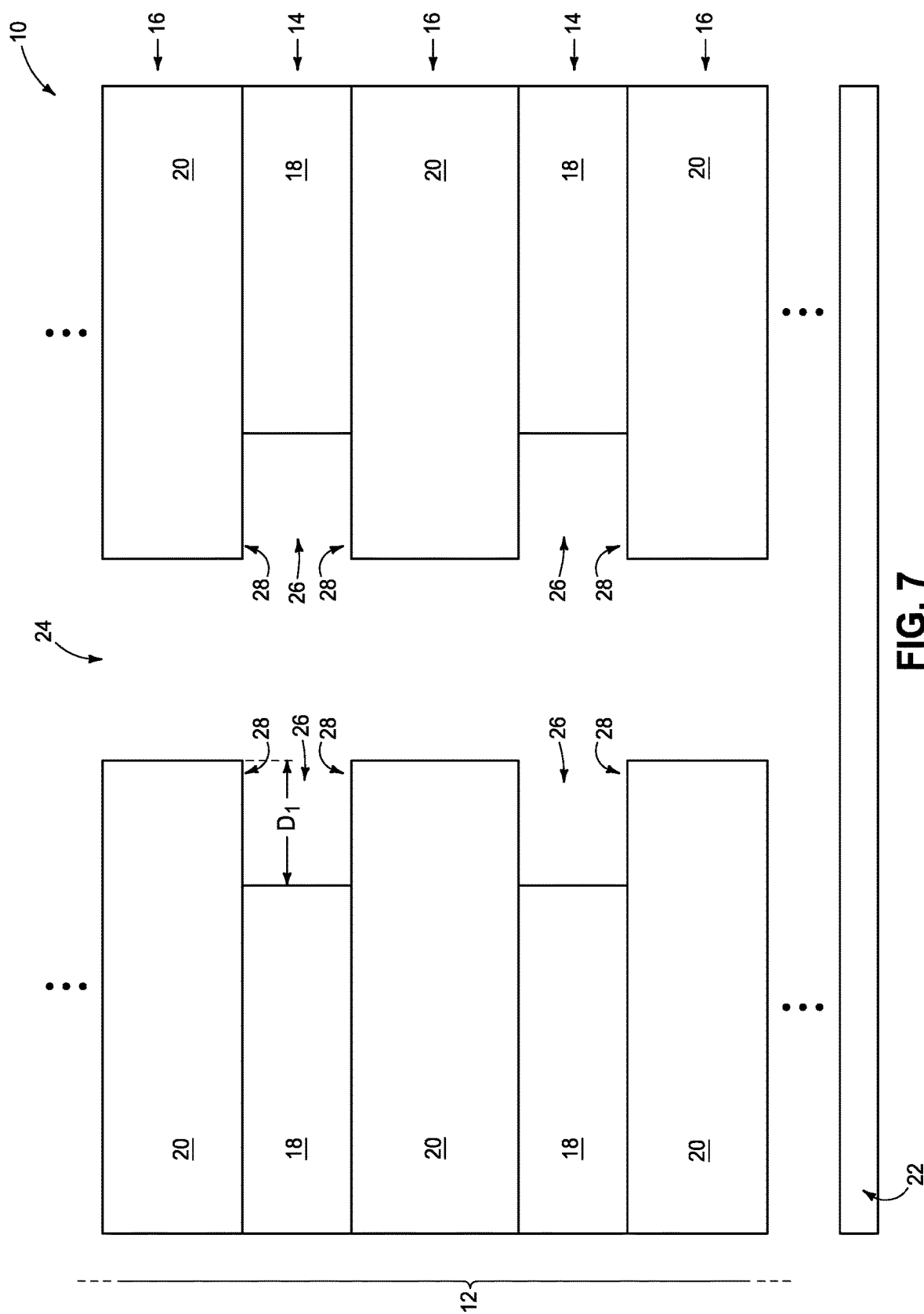
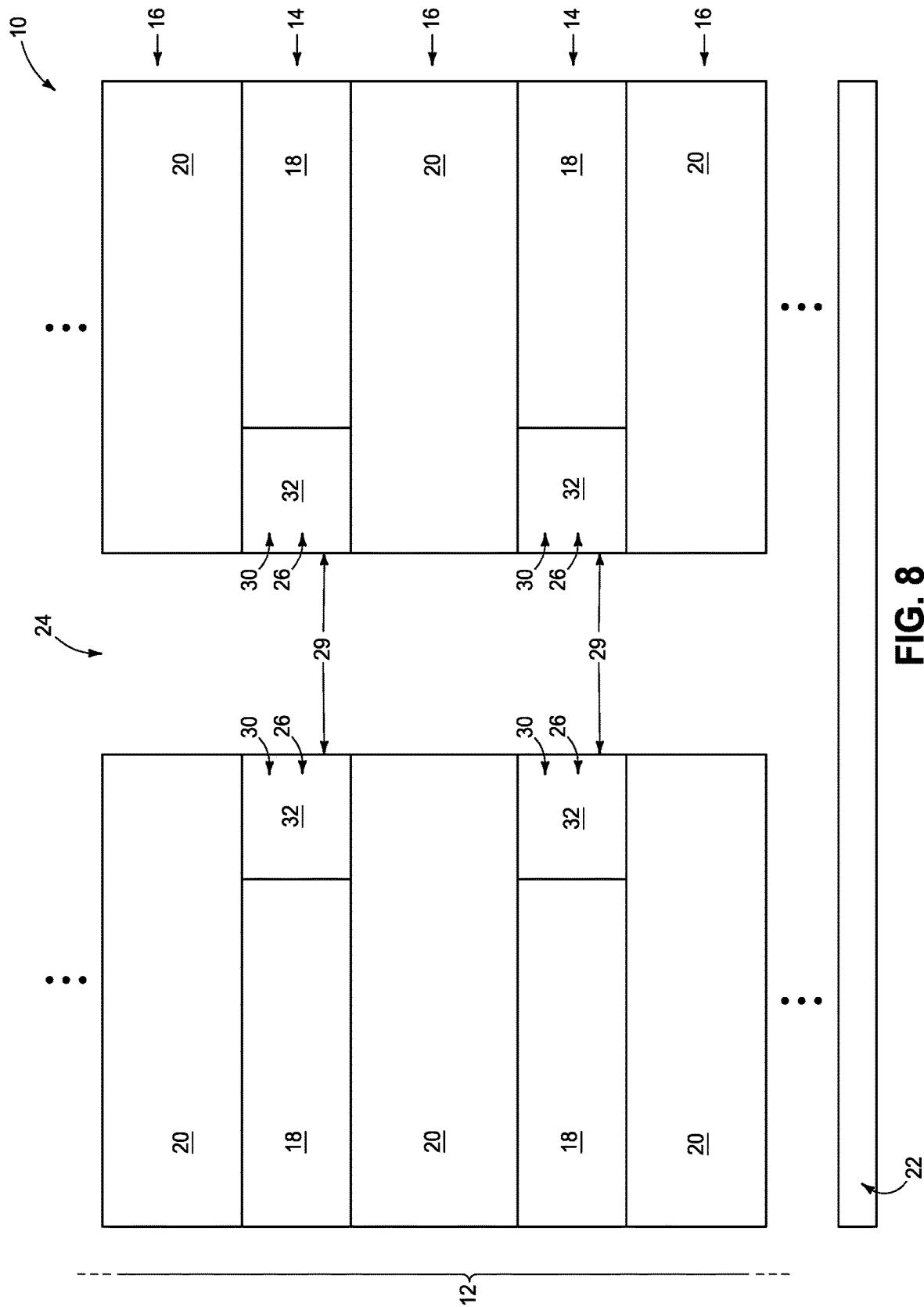
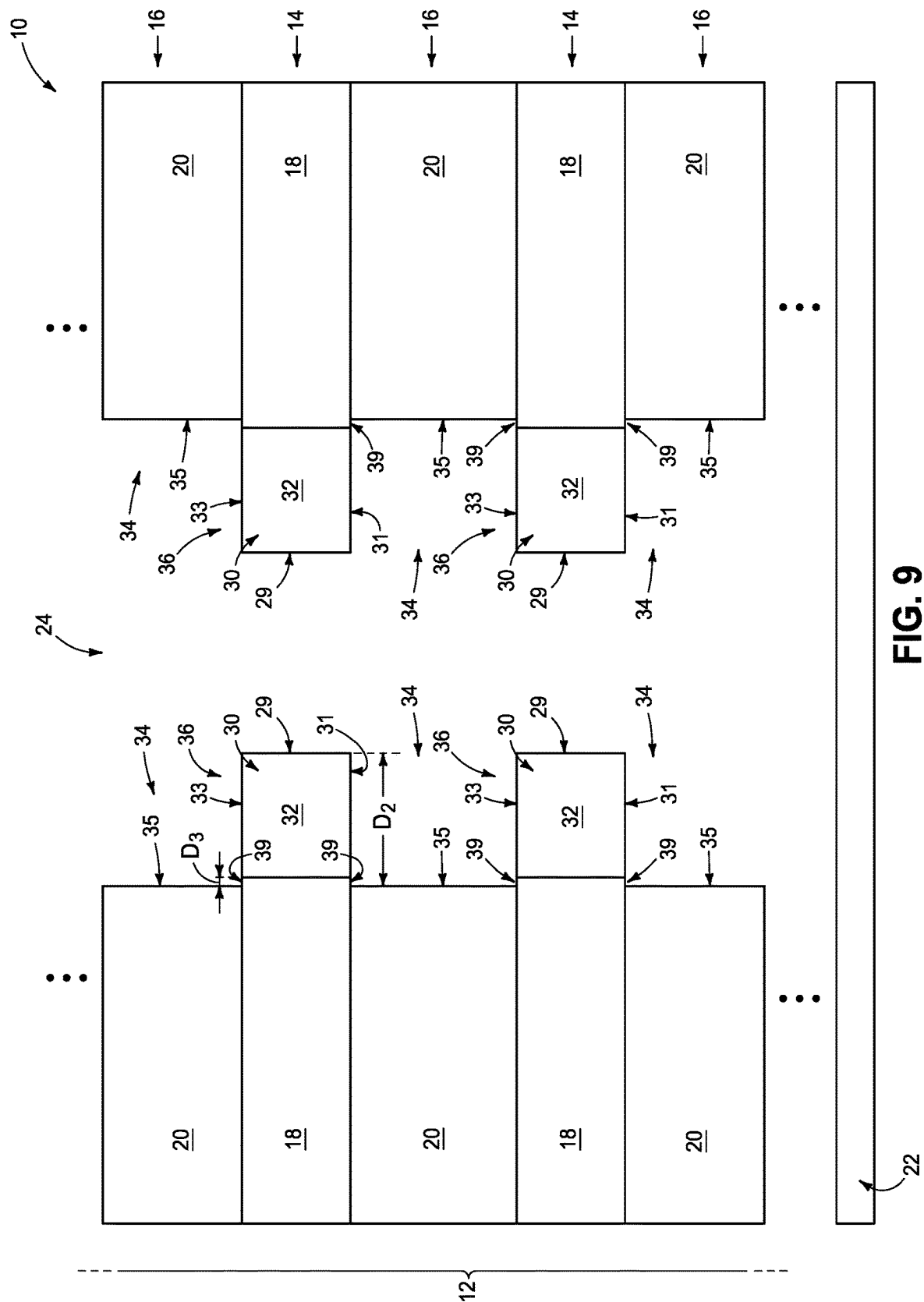
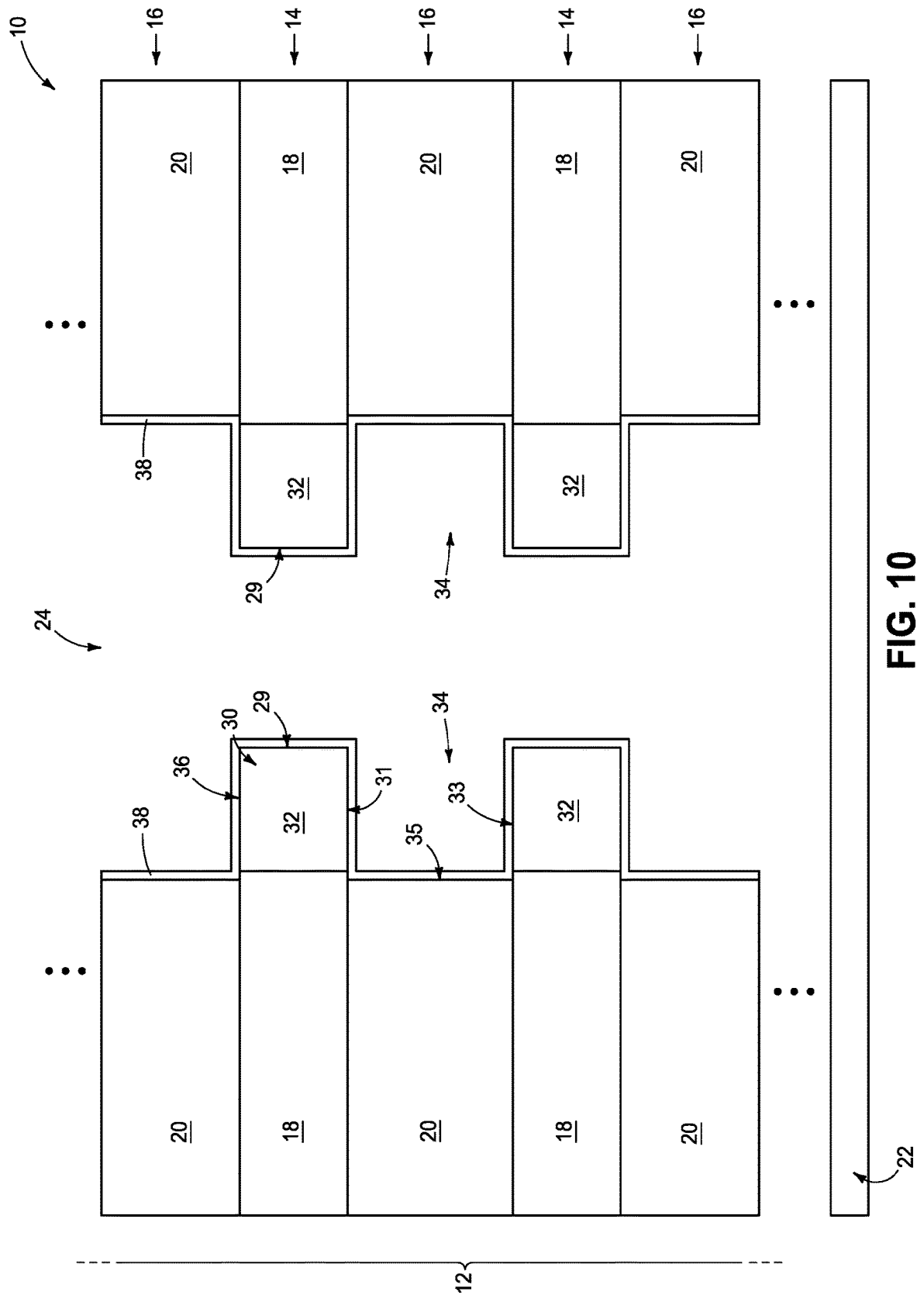


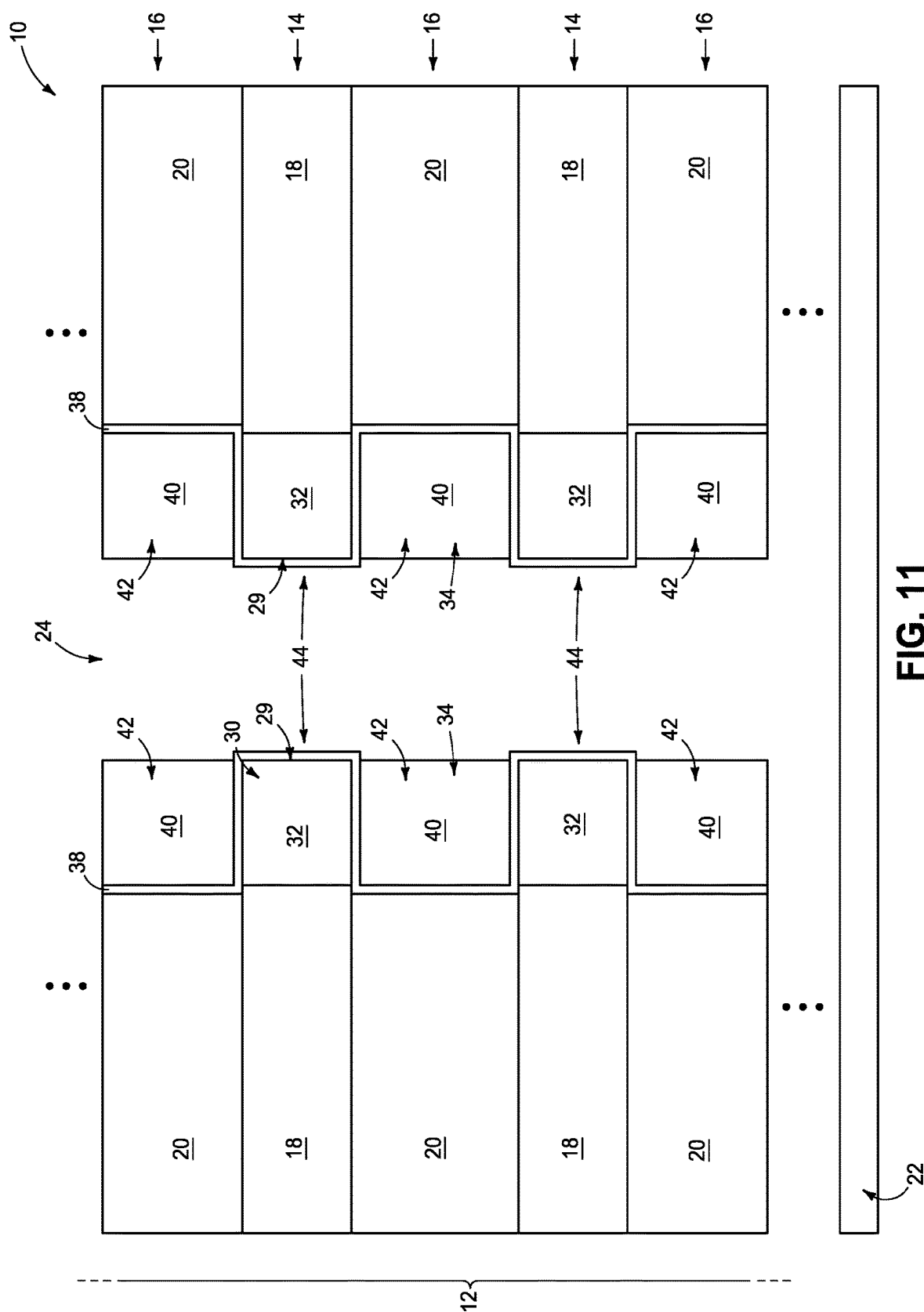
FIG. 6A

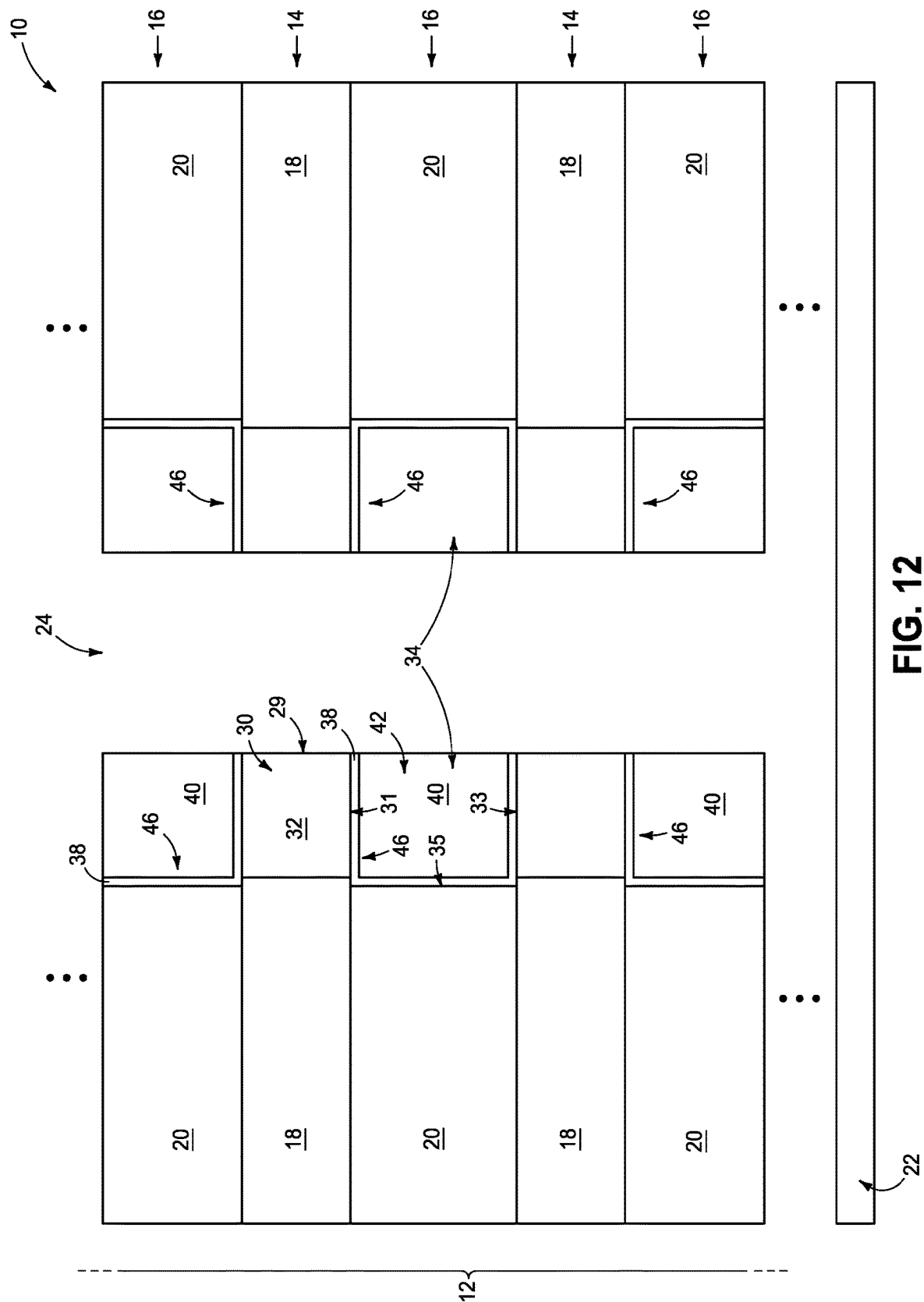


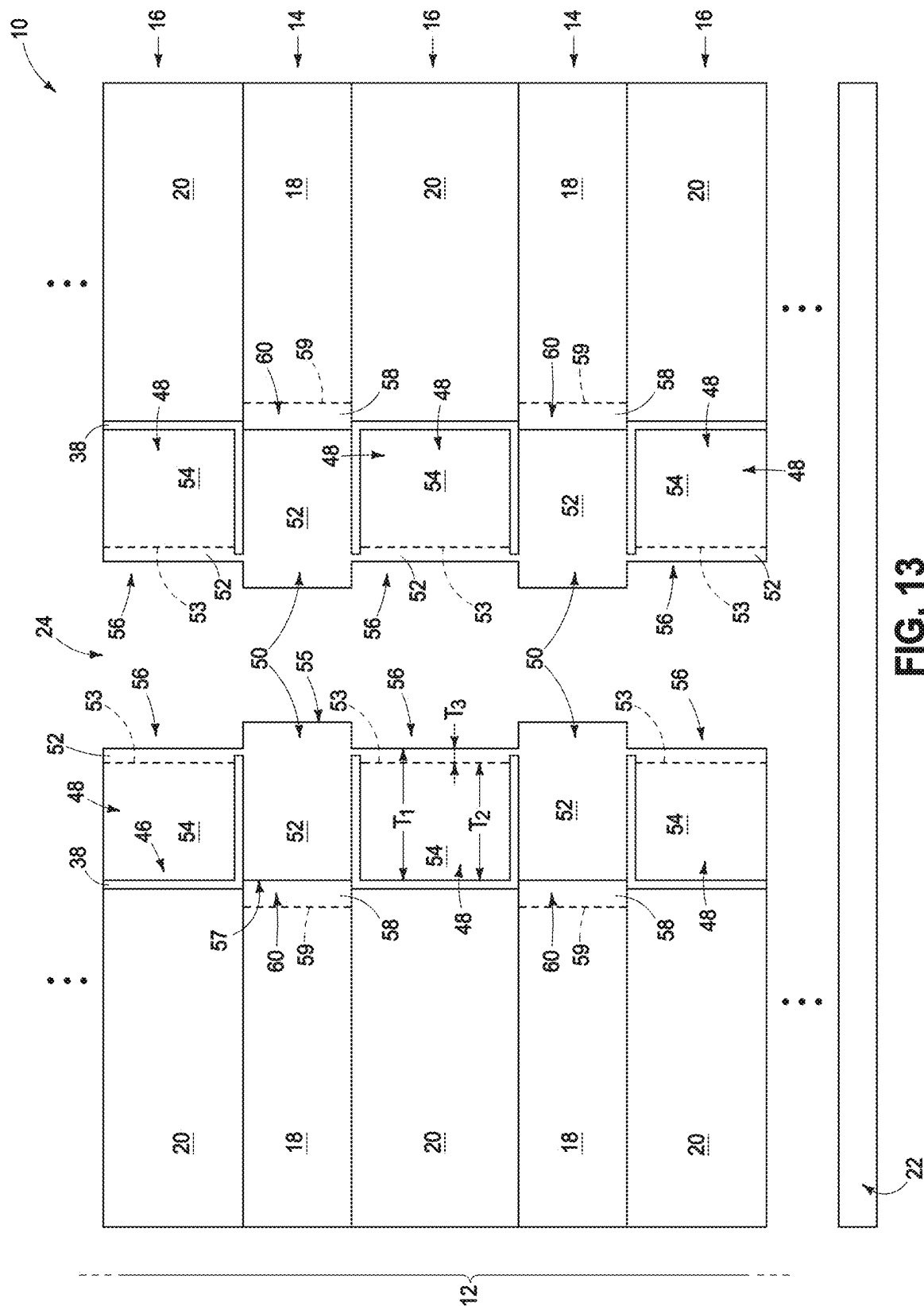


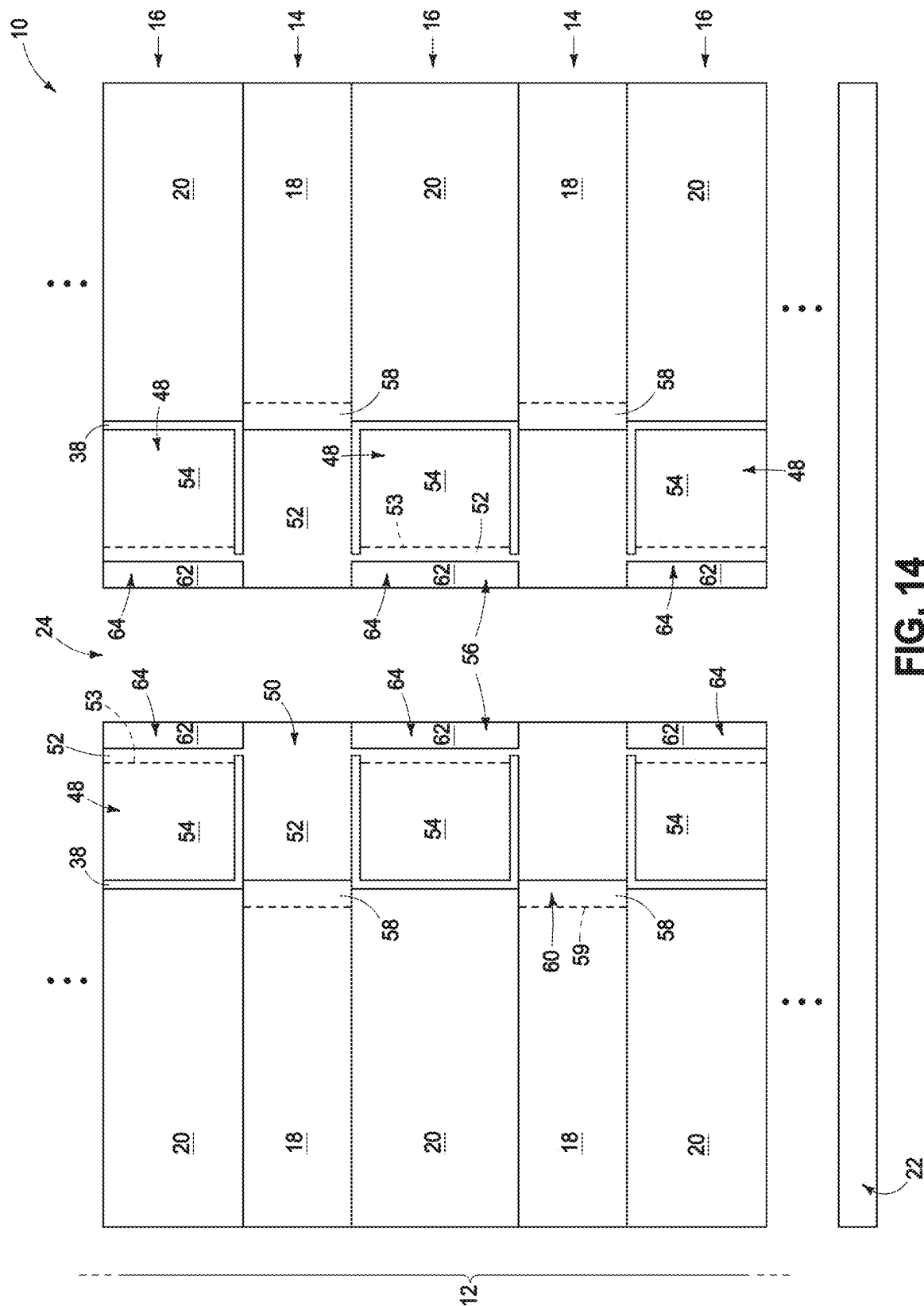


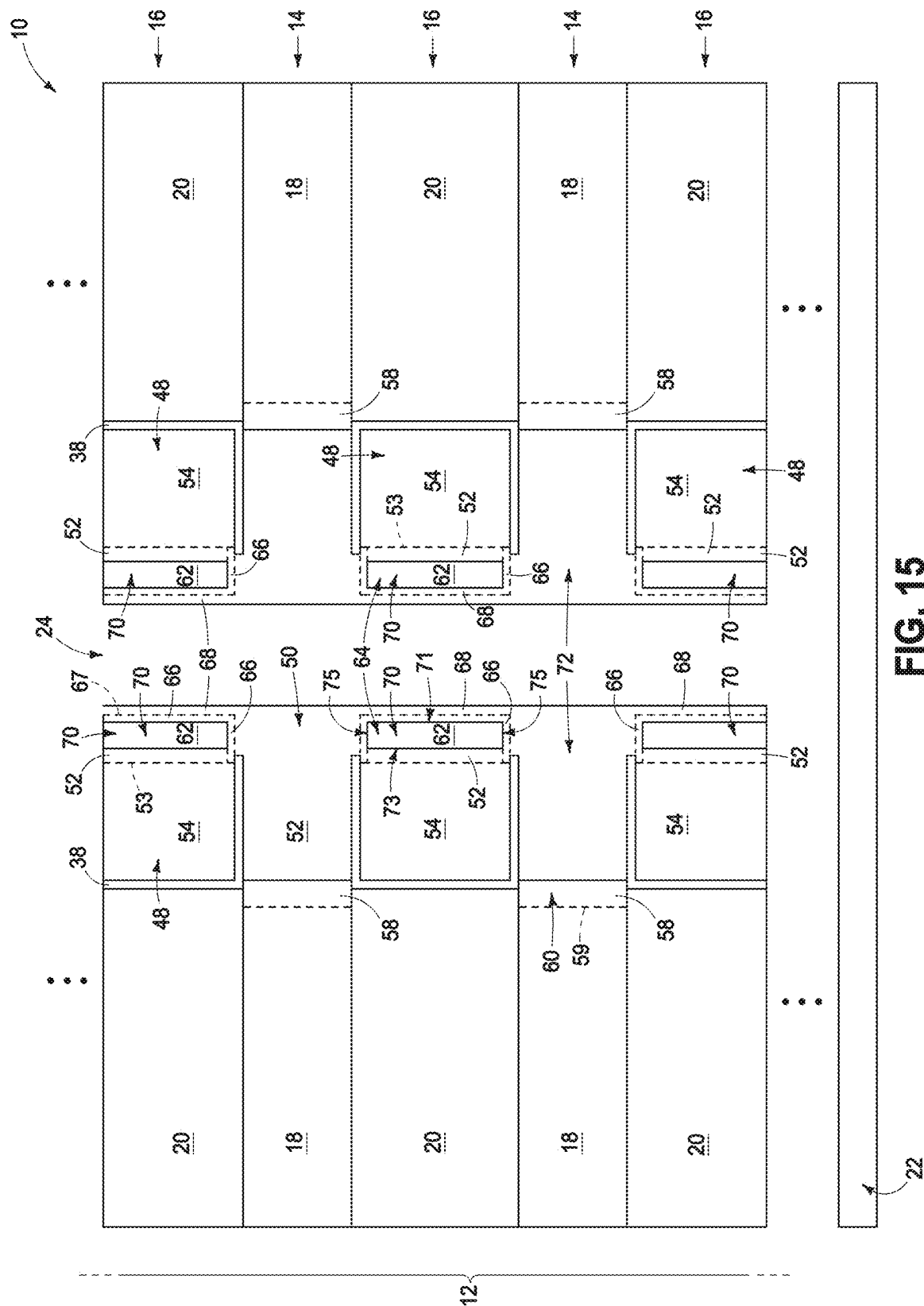


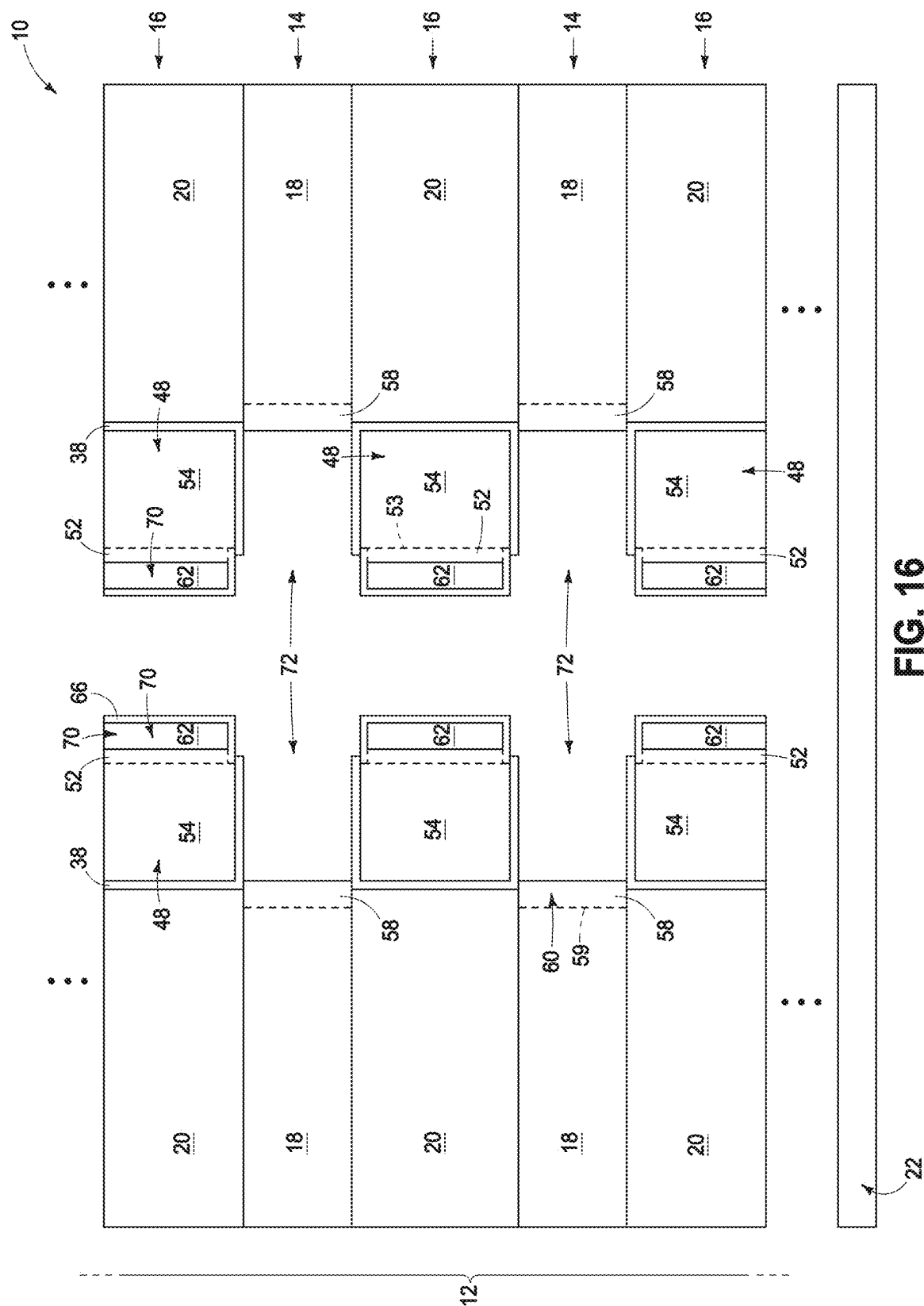


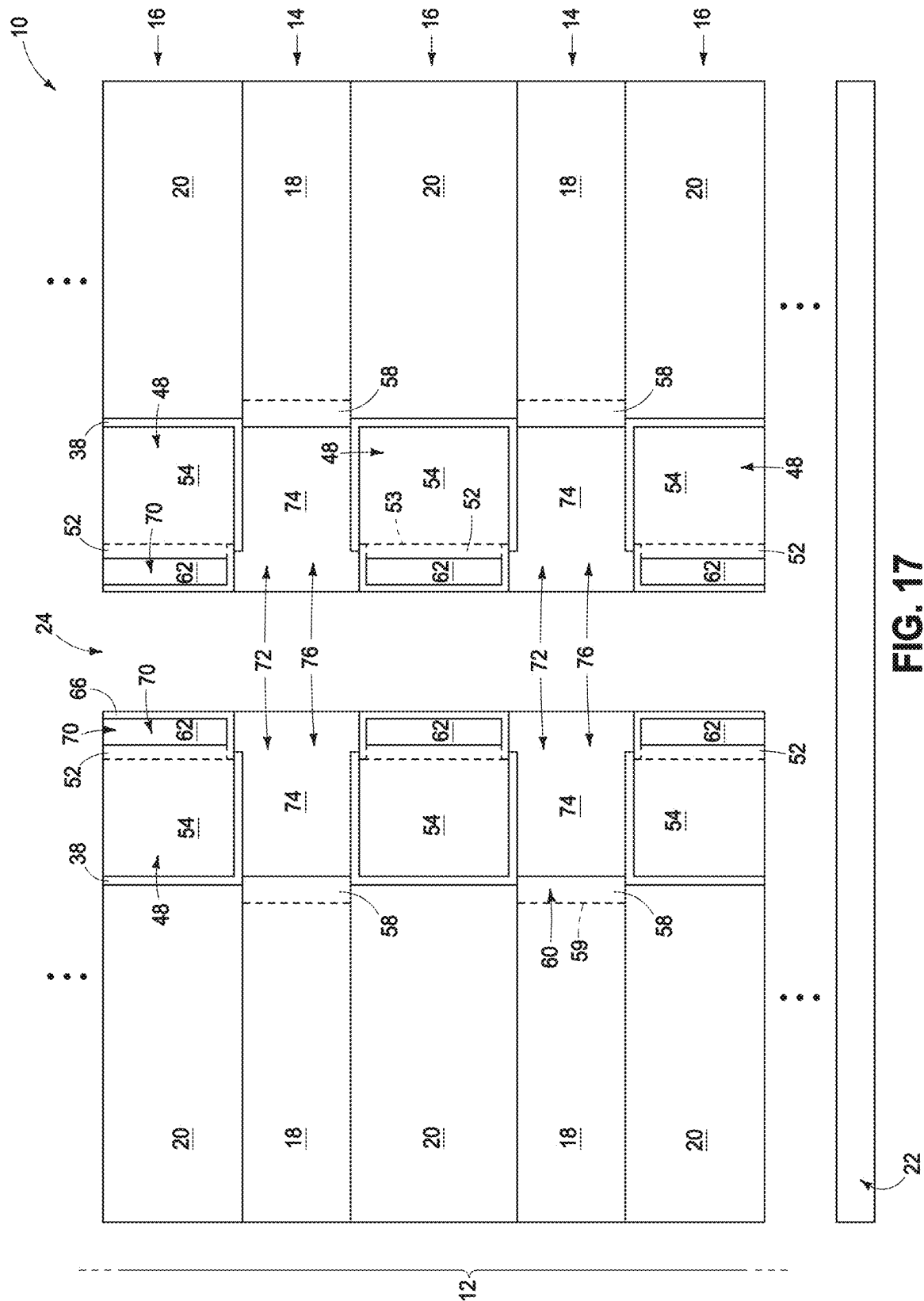


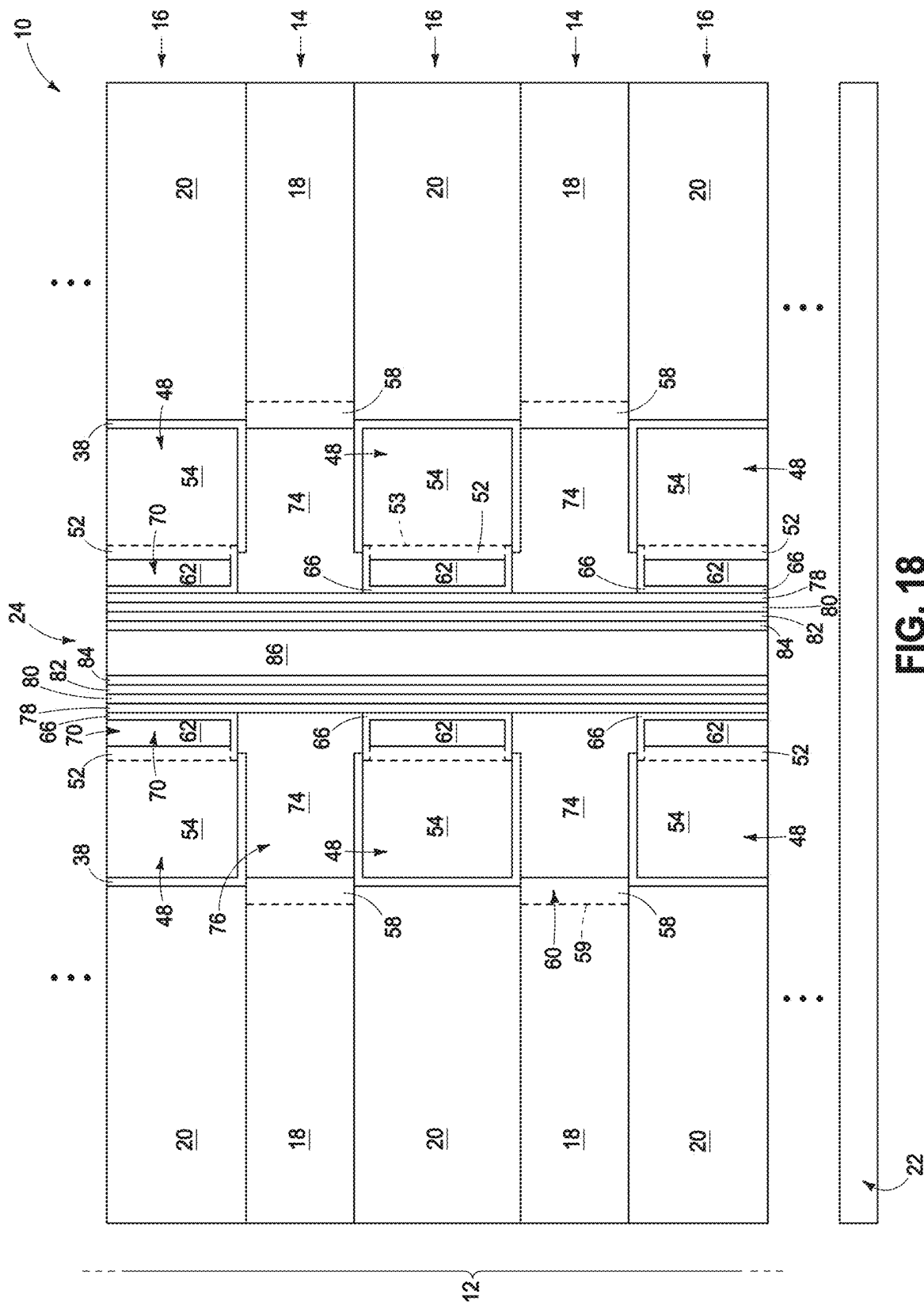


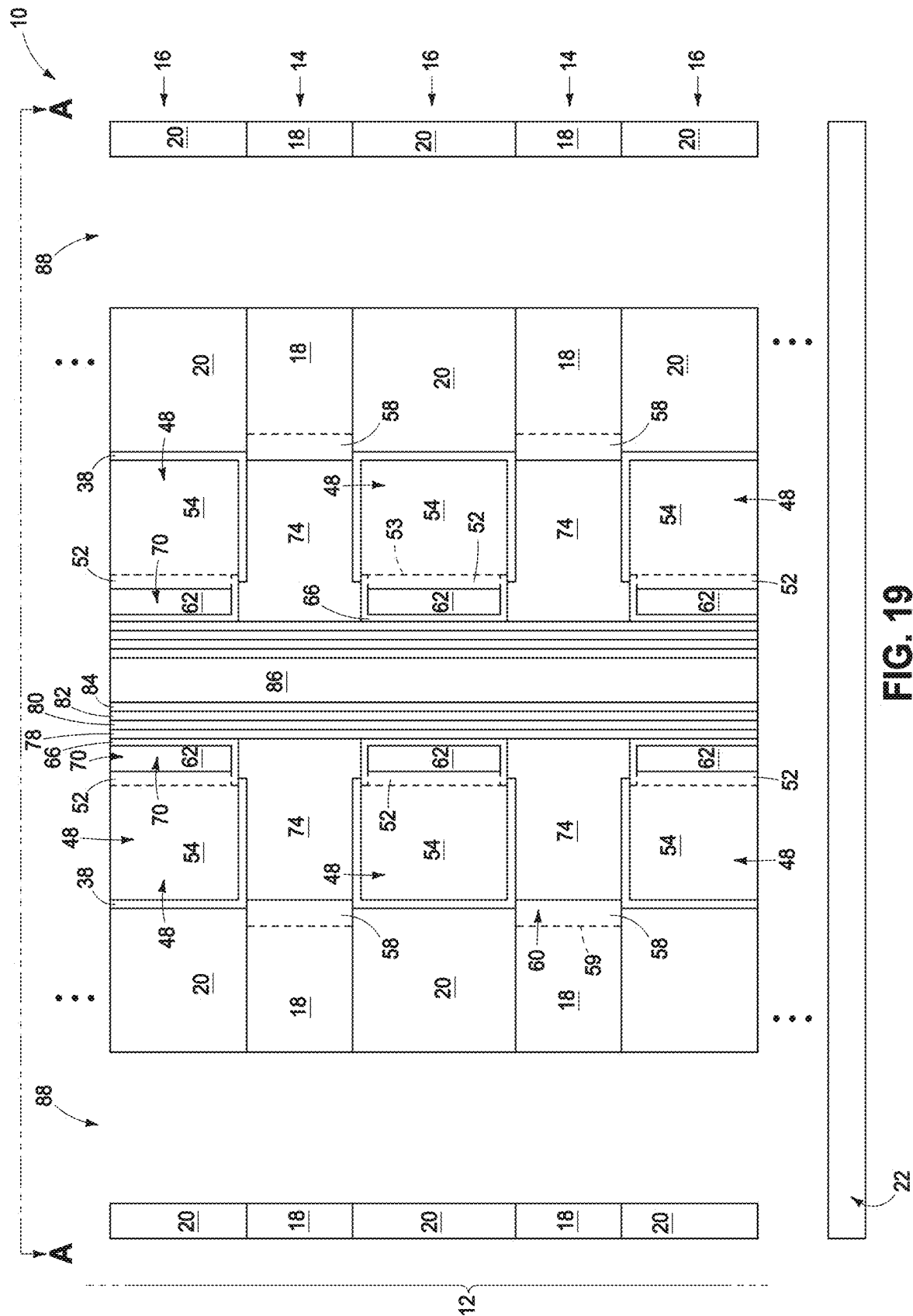












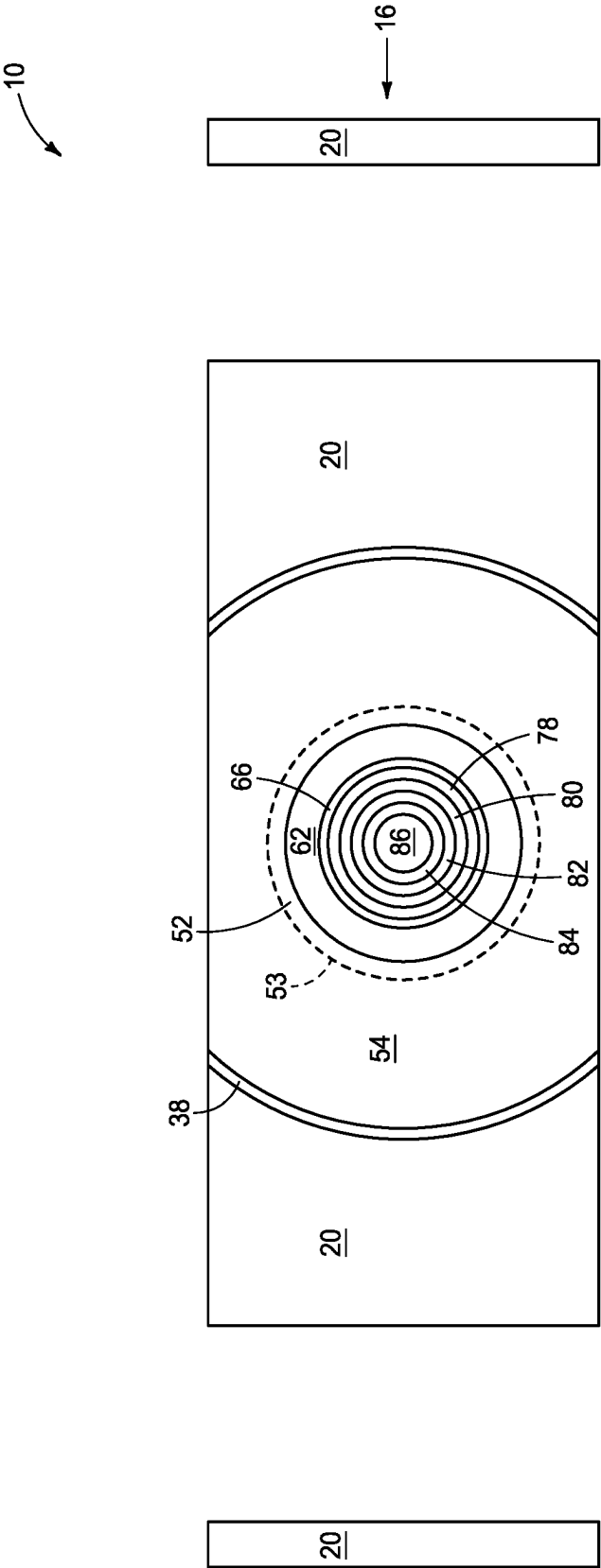


FIG. 19A

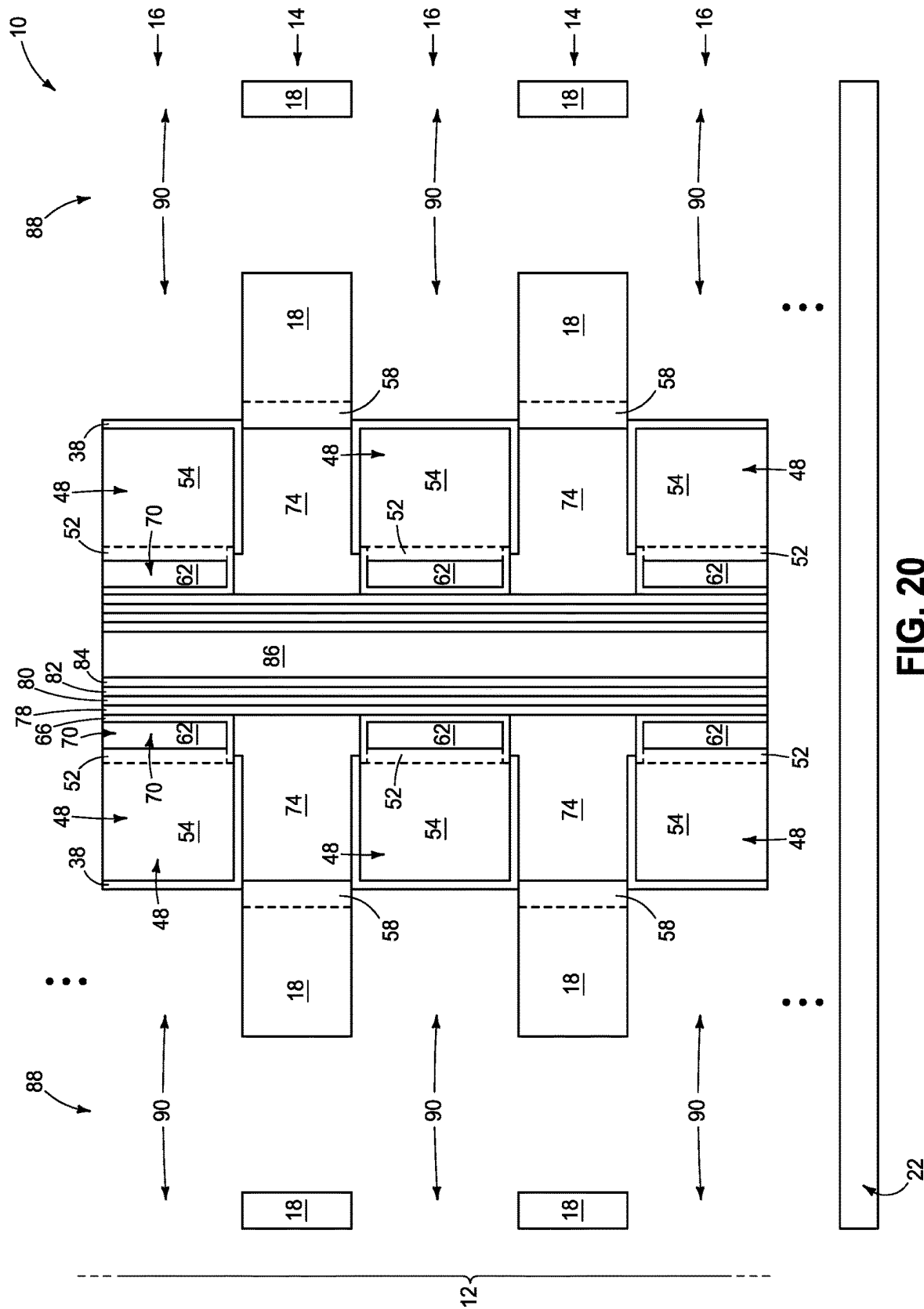


FIG. 20

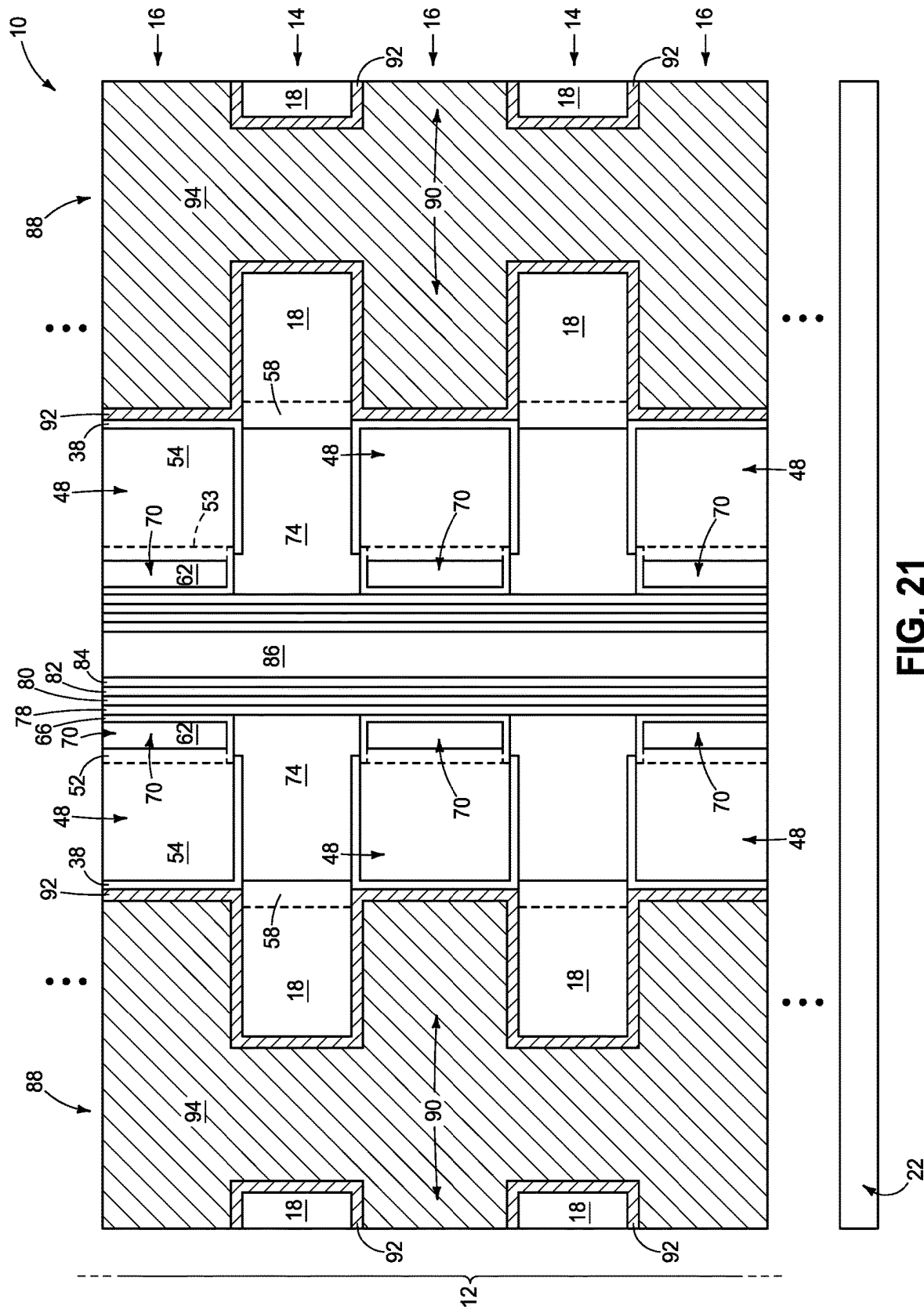


FIG. 21

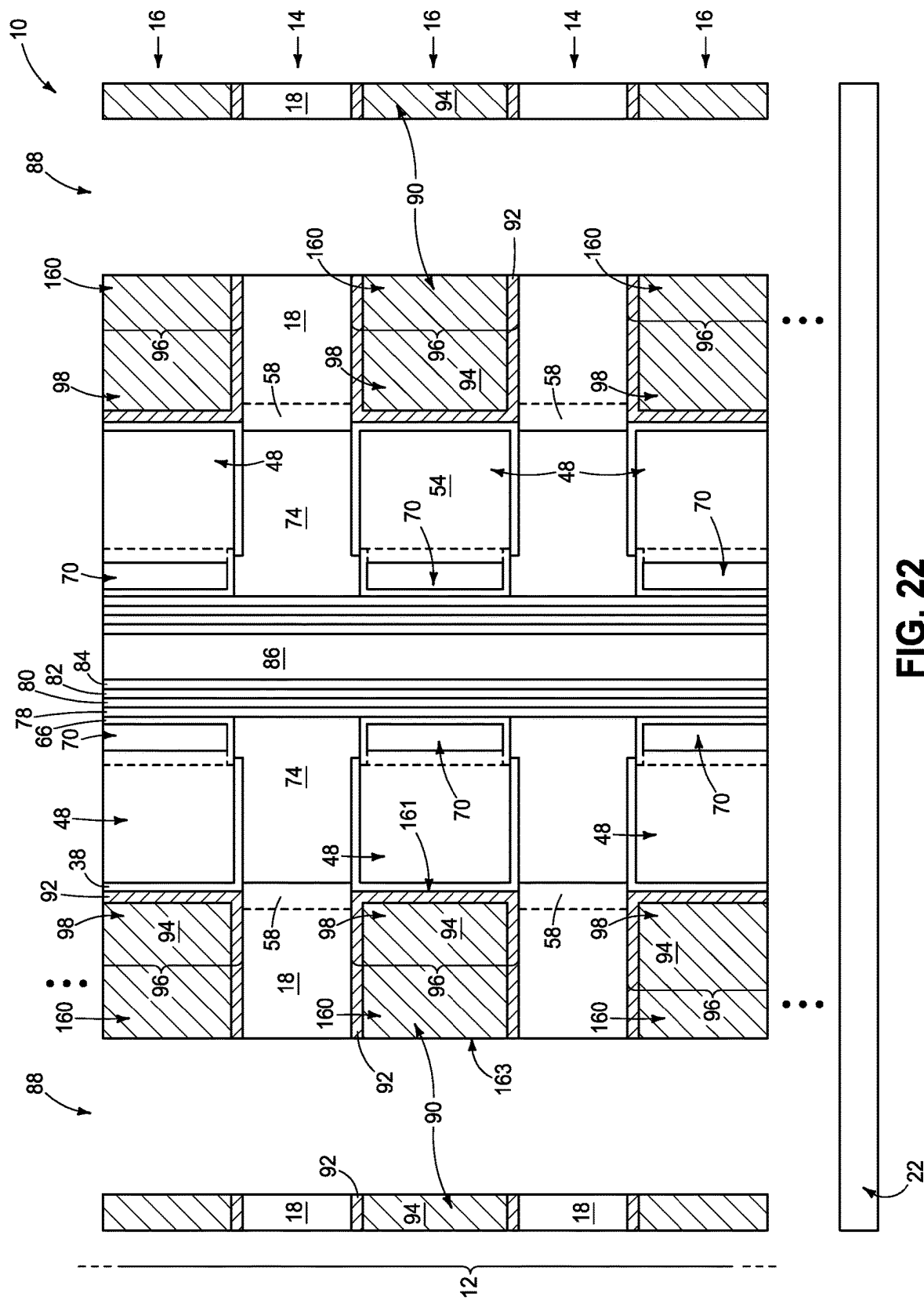
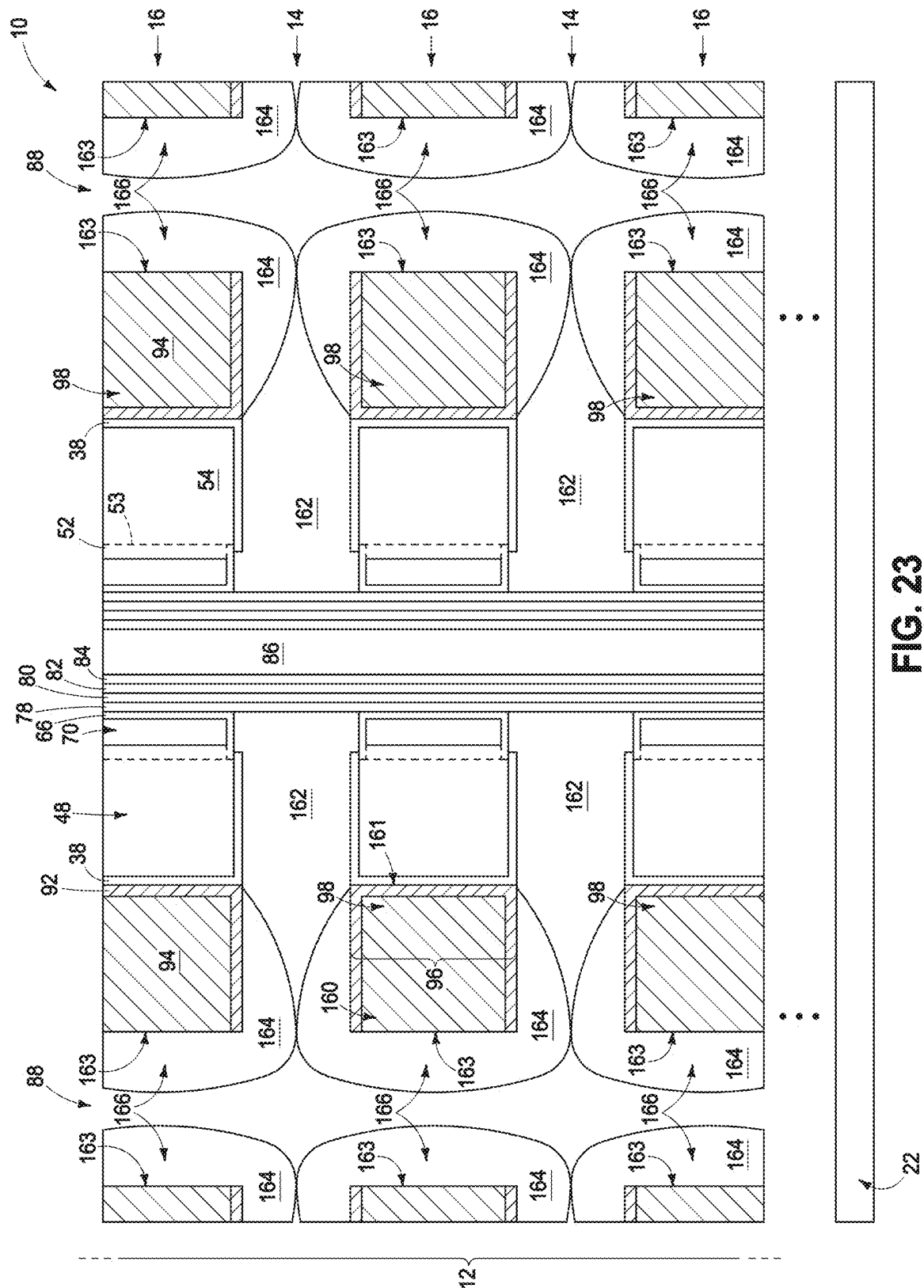
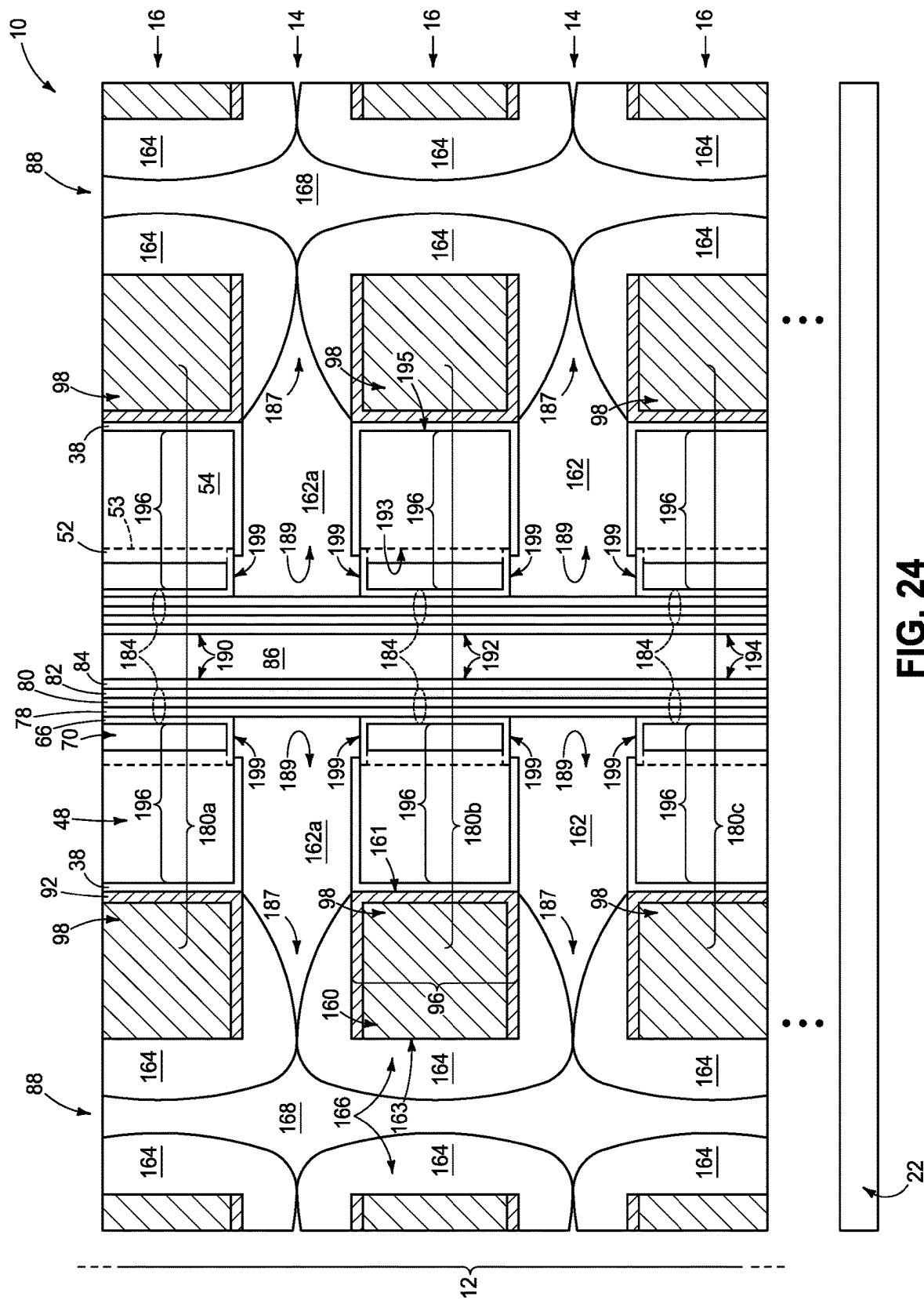
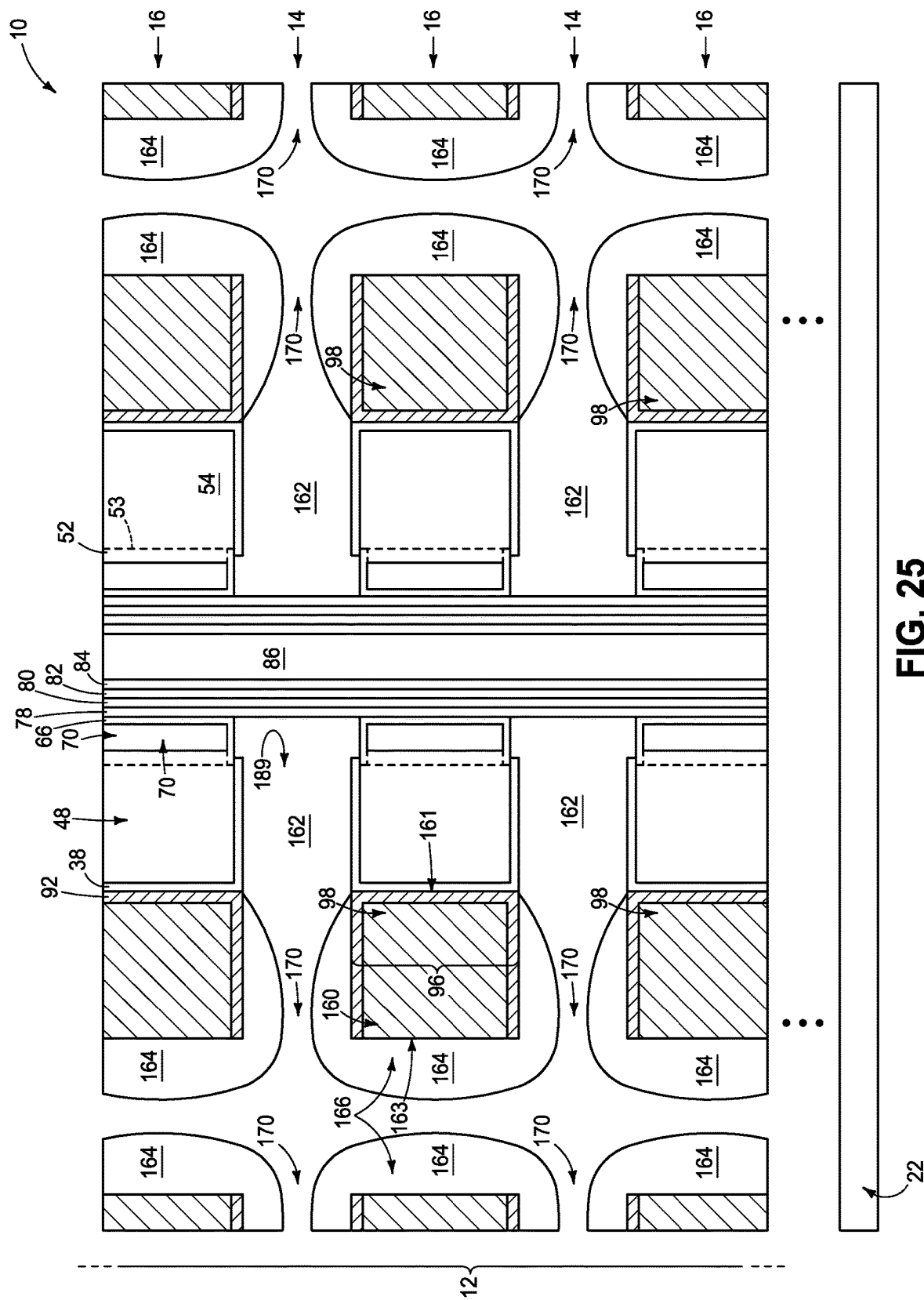
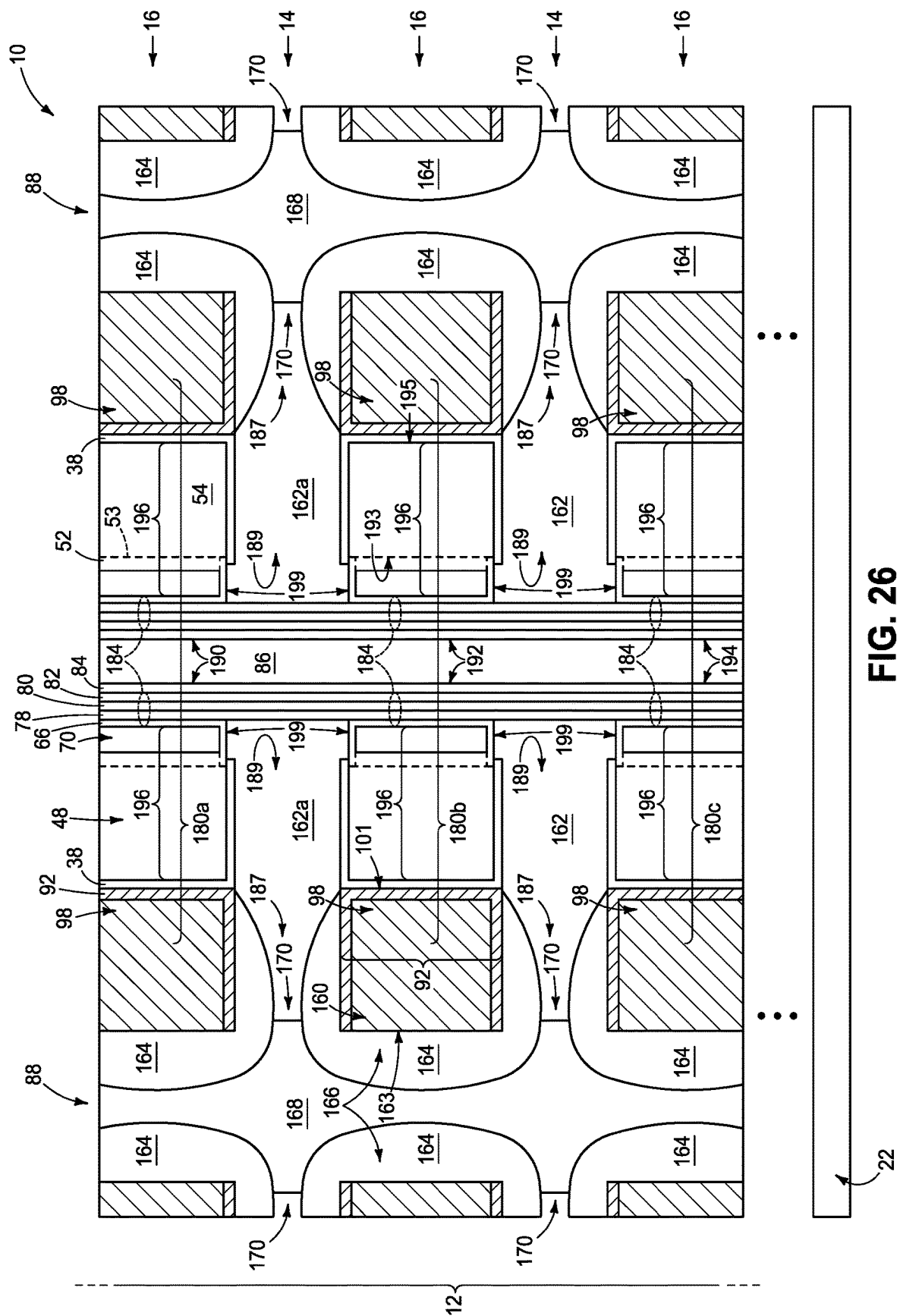


FIG. 22









MEMORY ARRAYS, AND METHODS OF FORMING MEMORY ARRAYS

RELATED PATENT DATA

This patent resulted from a continuation of U.S. patent application Ser. No. 16/674,823 filed Nov. 5, 2019, which is a divisional of U.S. patent application Ser. No. 15/948,639 filed Apr. 9, 2018, now U.S. Pat. No. 10,497,715, which claims priority to and the benefit of U.S. Provisional Application Ser. No. 62/610,657 filed Dec. 27, 2017, each of which is hereby incorporated by reference herein.

TECHNICAL FIELD

Memory arrays (e.g., NAND memory arrays), and methods of forming memory arrays.

BACKGROUND

Memory provides data storage for electronic systems. Flash memory is one type of memory, and has numerous uses in modern computers and devices. For instance, modern personal computers may have BIOS stored on a flash memory chip. As another example, it is becoming increasingly common for computers and other devices to utilize flash memory in solid state drives to replace conventional hard drives. As yet another example, flash memory is popular in wireless electronic devices because it enables manufacturers to support new communication protocols as they become standardized, and to provide the ability to remotely upgrade the devices for enhanced features.

NAND may be a basic architecture of flash memory, and may be configured to comprise vertically-stacked memory cells.

Before describing NAND specifically, it may be helpful to more generally describe the relationship of a memory array within an integrated arrangement. FIG. 1 shows a block diagram of a prior art device 100 which includes a memory array 102 having a plurality of memory cells 103 arranged in rows and columns along with access lines 104 (e.g., wordlines to conduct signals WL0 through WLn) and first data lines 106 (e.g., bitlines to conduct signals BL0 through BLn). Access lines 104 and first data lines 106 may be used to transfer information to and from the memory cells 103. A row decoder 107 and a column decoder 108 decode address signals A0 through AX on address lines 109 to determine which ones of the memory cells 103 are to be accessed. A sense amplifier circuit 115 operates to determine the values of information read from the memory cells 103. An I/O circuit 117 transfers values of information between the memory array 102 and input/output (I/O) lines 105. Signals DQ0 through DQN on the I/O lines 105 can represent values of information read from or to be written into the memory cells 103. Other devices can communicate with the device 100 through the I/O lines 105, the address lines 109, or the control lines 120. A memory control unit 118 is used to control memory operations to be performed on the memory cells 103, and utilizes signals on the control lines 120. The device 100 can receive supply voltage signals Vcc and Vss on a first supply line 130 and a second supply line 132, respectively. The device 100 includes a select circuit 140 and an input/output (I/O) circuit 117. The select circuit 140 can respond, via the I/O circuit 117, to signals CSEL1 through CSELn to select signals on the first data lines 106 and the second data lines 113 that can represent the values of information to be read from or to be programmed into the

memory cells 103. The column decoder 108 can selectively activate the CSEL1 through CSELn signals based on the A0 through AX address signals on the address lines 109. The select circuit 140 can select the signals on the first data lines 106 and the second data lines 113 to provide communication between the memory array 102 and the I/O circuit 117 during read and programming operations.

The memory array 102 of FIG. 1 may be a NAND memory array, and FIG. 2 shows a block diagram of a three-dimensional NAND memory device 200 which may be utilized for the memory array 102 of FIG. 1. The device 200 comprises a plurality of strings of charge-storage devices. In a first direction (Z-Z'), each string of charge-storage devices may comprise, for example, thirty-two charge-storage devices stacked over one another with each charge-storage device corresponding to one of, for example, thirty-two tiers (e.g., Tier0-Tier31). The charge-storage devices of a respective string may share a common channel region, such as one formed in a respective pillar of semiconductor material (e.g., polysilicon) about which the string of charge-storage devices is formed. In a second direction (X-X'), each first group of, for example, sixteen first groups of the plurality of strings may comprise, for example, eight strings sharing a plurality (e.g., thirty-two) of access lines (i.e., "global control gate (CG) lines", also known as wordlines, WLs). Each of the access lines may couple the charge-storage devices within a tier. The charge-storage devices coupled by the same access line (and thus corresponding to the same tier) may be logically grouped into, for example, two pages, such as P0/P32, P1/P33, P2/P34 and so on, when each charge-storage device comprises a cell capable of storing two bits of information. In a third direction (Y-Y'), each second group of, for example, eight second groups of the plurality of strings, may comprise sixteen strings coupled by a corresponding one of eight data lines. The size of a memory block may comprise 1,024 pages and total about 16 MB (e.g., 16 WLs×32 tiers×2 bits=1,024 pages/block, block size=1,024 pages×16 KB/page=16 MB). The number of the strings, tiers, access lines, data lines, first groups, second groups and/or pages may be greater or smaller than those shown in FIG. 2.

FIG. 3 shows a cross-sectional view of a memory block 300 of the 3D NAND memory device 200 of FIG. 2 in an X-X' direction, including fifteen strings of charge-storage devices in one of the sixteen first groups of strings described with respect to FIG. 2. The plurality of strings of the memory block 300 may be grouped into a plurality of subsets 310, 320, 330 (e.g., tile columns), such as tile column₁, tile column₂, and tile column_K, with each subset (e.g., tile column) comprising a "partial block" of the memory block 300. A global drain-side select gate (SGD) line 340 may be coupled to the SGDs of the plurality of strings. For example, the global SGD line 340 may be coupled to a plurality (e.g., three) of sub-SGD lines 342, 344, 346 with each sub-SGD line corresponding to a respective subset (e.g., tile column), via a corresponding one of a plurality (e.g., three) of sub-SGD drivers 332, 334, 336. Each of the sub-SGD drivers 332, 334, 336 may concurrently couple or cut off the SGDs of the strings of a corresponding partial block (e.g., tile column) independently of those of other partial blocks. A global source-side select gate (SGS) line 360 may be coupled to the SGSs of the plurality of strings. For example, the global SGS line 360 may be coupled to a plurality of sub-SGS lines 362, 364, 366 with each sub-SGS line corresponding to the respective subset (e.g., tile column), via a corresponding one of a plurality of sub-SGS drivers 322, 324, 326. Each of the

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sub-SGS drivers **322**, **324**, **326** may concurrently couple or cut off the SGSs of the strings of a corresponding partial block (e.g., tile column) independently of those of other partial blocks. A global access line (e.g., a global CG line) **350** may couple the charge-storage devices corresponding to the respective tier of each of the plurality of strings. Each global CG line (e.g., the global CG line **350**) may be coupled to a plurality of sub-access lines (e.g., sub-CG lines) **352**, **354**, **356** via a corresponding one of a plurality of sub-string drivers **312**, **314** and **316**. Each of the sub-string drivers may concurrently couple or cut off the charge-storage devices corresponding to the respective partial block and/or tier independently of those of other partial blocks and/or other tiers. The charge-storage devices corresponding to the respective subset (e.g., partial block) and the respective tier may comprise a "partial tier" (e.g., a single "tile") of charge-storage devices. The strings corresponding to the respective subset (e.g., partial block) may be coupled to a corresponding one of sub-sources **372**, **374** and **376** (e.g., "tile source") with each sub-source being coupled to a respective power source.

The NAND memory device **200** is alternatively described with reference to a schematic illustration of FIG. 4.

The memory array **200** includes wordlines **202₁** to **202_N**, and bitlines **228₁** to **228_M**.

The memory array **200** also includes NAND strings **206₁** to **206_M**. Each NAND string includes charge-storage transistors **208₁** to **208_N**. The charge-storage transistors may use floating gate material (e.g., polysilicon) to store charge, or may use charge-trapping material (such as, for example, silicon nitride, metallic nanodots, etc.) to store charge.

The charge-storage transistors **208** are located at intersections of wordlines **202** and strings **206**. The charge-storage transistors **208** represent non-volatile memory cells for storage of data. The charge-storage transistors **208** of each NAND string **206** are connected in series source-to-drain between a source-select device (e.g., source-side select gate, SGS) **210** and a drain-select device (e.g., drain-side select gate, SGD) **212**. Each source-select device **210** is located at an intersection of a string **206** and a source-select line **214**, while each drain-select device **212** is located at an intersection of a string **206** and a drain-select line **215**. The select devices **210** and **212** may be any suitable access devices, and are generically illustrated with boxes in FIG. 1.

A source of each source-select device **210** is connected to a common source line **216**. The drain of each source-select device **210** is connected to the source of the first charge-storage transistor **208** of the corresponding NAND string **206**. For example, the drain of source-select device **210₁** is connected to the source of charge-storage transistor **208₁** of the corresponding NAND string **206₁**. The source-select devices **210** are connected to source-select line **214**.

The drain of each drain-select device **212** is connected to a bitline (i.e., digit line) **228** at a drain contact. For example, the drain of drain-select device **212₁** is connected to the bitline **228₁**. The source of each drain-select device **212** is connected to the drain of the last charge-storage transistor **208** of the corresponding NAND string **206**. For example, the source of drain-select device **212₁** is connected to the drain of charge-storage transistor **208_N** of the corresponding NAND string **206₁**.

The charge-storage transistors **208** include a source **230**, a drain **232**, a charge-storage region **234**, and a control gate **236**. The charge-storage transistors **208** have their control gates **236** coupled to a wordline **202**. A column of the charge-storage transistors **208** are those transistors within a NAND string **206** coupled to a given bitline **228**. A row of

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the charge-storage transistors **208** are those transistors commonly coupled to a given wordline **202**.

It would be desirable to develop improved memory cell designs, improved memory array architecture (e.g., improved NAND architecture), and methods for fabricating the improved memory cells and improved memory array architectures.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows a block diagram of a prior art memory device having a memory array with memory cells.

FIG. 2 shows a schematic diagram of the prior art memory array of FIG. 1 in the form of a 3D NAND memory device.

FIG. 3 shows a cross-sectional view of the prior art 3D NAND memory device of FIG. 2 in an X-X' direction.

FIG. 4 is a schematic diagram of a prior art NAND memory array.

FIGS. 5-24 are diagrammatic cross-sectional views of regions of a construction at example process steps of an example method for fabricating an example assembly.

FIGS. 6A and 19A are diagrammatic top views along the lines A-A of FIGS. 6 and 19, respectively.

FIGS. 25 and 26 are diagrammatic cross-sectional views of regions of the construction of FIG. 22 at example process steps that may follow the example process stage of FIG. 22 in an example method different from that described with reference to FIGS. 23 and 24.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENTS

Some embodiments include memory arrays having low-density silicon dioxide wrapping around ends of wordline levels. Some embodiments include memory cells having charge-blocking regions which comprise silicon oxynitride. The charge-blocking regions may additionally comprise silicon dioxide. Some embodiments include memory arrays (e.g., NAND memory arrays) having vertically-stacked memory cells, and having voids between vertically-adjacent memory cells. Some embodiments include methods of forming memory cells, and memory arrays. Example methods are described with reference to FIGS. 5-26, and example architectures are described with reference to FIGS. 24 and 26.

Referring to FIG. 5, a construction (i.e., assembly, architecture, etc.) **10** includes a stack **12** of alternating first and second levels **14** and **16**. The first levels **14** comprise first material **18**, and the second levels **16** comprise second material **20**. The first and second materials **18** and **20** may be any suitable materials. In some embodiments, the first material **18** may comprise, consist essentially of, or consist of silicon nitride; and the second material **20** may comprise, consist essentially of, or consist of silicon dioxide.

The levels **14** and **16** may be of any suitable thicknesses; and may be the same thickness as one another, or different thicknesses relative to one another. In some embodiments, the levels **14** and **16** may have vertical thicknesses within a range of from about 10 nanometers (nm) to about 400 nm. In some embodiments, the second levels **16** may be thicker than the first levels **14**. For instance, in some embodiments the second levels **16** may have thicknesses within a range of from about 20 nm to about 40 nm, and the first levels **14** may have thicknesses within a range of from about 15 nm to about 30 nm.

Some of the material **20** of the second levels **16** is ultimately replaced with conductive material of memory cell gates. Accordingly, the levels **16** may ultimately correspond

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to memory cell levels of a NAND configuration. The NAND configuration will include strings of memory cells (i.e., NAND strings), with the number of memory cells in the strings being determined by the number of vertically-stacked levels **16**. The NAND strings may comprise any suitable number of memory cell levels. For instance, the NAND strings may have 8 memory cell levels, 16 and memory cell levels, 32 memory cell levels, 64 memory cell levels, 512 memory cell levels, 1024 memory cell levels, etc. A portion or portions of the cell levels may be used as select gate(s). The vertical stack **12** is shown to extend outwardly beyond the illustrated region of the stack to indicate that there may be more vertically-stacked levels than those specifically illustrated in the diagram of FIG. 5.

The stack **12** is shown to be supported over a base **22**. The base **22** may comprise semiconductor material; and may, for example, comprise, consist essentially of, or consist of monocrystalline silicon. The base **22** may be referred to as a semiconductor substrate. The term “semiconductor substrate” means any construction comprising semiconductive material, including, but not limited to, bulk semiconductive materials such as a semiconductive wafer (either alone or in assemblies comprising other materials), and semiconductive material layers (either alone or in assemblies comprising other materials). The term “substrate” refers to any supporting structure, including, but not limited to, the semiconductor substrates described above. In some applications, the base **22** may correspond to a semiconductor substrate containing one or more materials associated with integrated circuit fabrication. Such materials may include, for example, one or more of refractory metal materials, barrier materials, diffusion materials, insulator materials, etc.

A space is provided between the stack **12** and the base **22** to indicate that other components and materials may be provided between the stack **12** and the base **22**. Such other components and materials may comprise additional levels of the stack, a source line level, source-side select gates (SGSs), etc.

Referring to FIG. 6, an opening **24** is formed through the stack **12**. The opening is ultimately utilized for fabricating channel material pillars associated with vertically-stacked memory cells of a memory array, and in some embodiments may be referred to as a pillar opening. The opening **24** may have any suitable configuration when viewed from above; and in some example embodiments may be circular, elliptical, polygonal, etc. FIG. 6A shows a top view of a portion of the top level **16** of the illustrated region of construction **10**, and illustrates an example configuration in which the opening **24** is circular-shaped when viewed from above. In some embodiments, the opening **24** may be referred to as a first opening in order to distinguish it from other openings formed at later process stages. The pillar opening **24** may be representative of a large plurality of substantially identical openings formed across the base **22** at the processing stage of FIG. 6 (with the term “substantially identical” meaning identical to within reasonable tolerances of fabrication and measurement).

Referring to FIG. 7, the material **18** of the first levels **14** is recessed along the opening **24** to form gaps (i.e., cavities) **26**. The gaps **26** may be referred to as first gaps to distinguish them from other gaps formed at subsequent process stages.

In some embodiments, the material **18** of the first levels **14** may comprise, consist essentially of, or consist of silicon nitride; and the material **20** of the second levels **16** may comprise, consist essentially of, or consist of silicon dioxide. In such embodiments, the material **18** may be selectively

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etched relative to the material **20** utilizing phosphoric acid. The term “selective etching” means that a material is removed faster than another material, and includes, but is not limited to, etching processes which are 100% selective for one material relative to another.

The first gaps **26** are vertically between segments **28** of the material **20** of the second levels **16**.

The first gaps extend into the first levels **14** to a depth D1. Such depth may be any suitable depth, and in some embodiments will be within a range of from about 10 nm to about 20 nm.

Referring to FIG. 8, spacing structures **30** are formed within the first gaps **26**. The spacing structures **30** comprise material **32**. Such material may comprise any suitable composition(s); and in some embodiments may comprise silicon. For instance, in some example embodiments, the material **32** may comprise, consist essentially of, or consist of polycrystalline silicon. The spacing structures **30** comprise outer edges **29** along the opening **24**.

The material **32** may be formed within the gaps **26** with any suitable processing. For instance, in some embodiments material **32** may be deposited through the opening **24** and into the gaps **26**; resulting in the gaps **26** being filled with the material **32** and the opening **24** being at least partially filled with the material **32**. Subsequently, an etch may be utilized to remove excess material **32** from within the opening **24**, while leaving the material **32** within the gaps **26**. The etch may utilize any suitable etchant and etching conditions. In some example embodiments, the etch will utilize tetramethylammonium hydroxide (TMAH).

Referring to FIG. 9, the material **20** of the second levels **16** is recessed along the opening **24** to form second gaps **34**. In some embodiments, the material **20** of the second levels **16** may comprise, consist essentially of, or consist of silicon dioxide; and the material **32** of the spacing structures **30** may comprise, consist essentially of, or consist of polycrystalline silicon. In such embodiments, the material **20** may be selectively etched relative to the material **32** utilizing a buffered oxide etch (e.g., an etch utilizing a hydrofluoric acid and a buffering agent, such as ammonium fluoride). The gaps **34** are vertically between segments **36** of the spacing structures **30**. Remaining portions of the second material **20** are behind the gaps **34** (and in some embodiments may be referred to as being along the gaps **34**).

The second gaps **34** extend into the first levels **14** to a depth D2. Such depth may be any suitable depth, and in some embodiments will be within a range of from about 10 nm to about 30 nm.

In some embodiments, the second gaps **34** may be each considered to comprise an upper periphery (or upper peripheral surface) **31**, a lower periphery (or lower peripheral surface) **33**, and an inner periphery (or inner peripheral surface **35**); with the inner periphery extending between the upper periphery and the lower periphery.

In the illustrated embodiment, regions **39** of the first material **18** are exposed along back portions of the second gaps **34**. The exposed regions are along a depth D3 at the back of the second gaps **34**. In some embodiments, the depth D3 may be within a range of from about 5 angstroms (Å) to about 70 Å. In alternative embodiments, regions of the first material **18** are not exposed along the back portions of the second gaps.

Referring to FIG. 10, a layer of dielectric barrier material **38** is formed to extend conformally along the peripheral surfaces **31**, **33** and **35** within the gaps **34**; and to extend along the outer edges **29** of the spacing structures **30**. The dielectric barrier material **38** may comprise any suitable

composition(s); and in some embodiments may comprise one or more high-k materials (with the term high-k meaning a dielectric constant greater than that of silicon dioxide). Example compositions which may be incorporated into the dielectric barrier material are hafnium oxide, zirconium oxide, aluminum oxide, hafnium silicate, zirconium silicate, titanium oxide, gadolinium oxide, niobium oxide, tantalum oxide, etc.

The dielectric barrier material **38** narrows the second gaps **34**.

Referring to FIG. **11**, a material **40** is formed within the narrowed gaps **34**. In some embodiments, the material **40** may comprise, consist essentially of, or consist of silicon nitride. The material **40** may be formed in the shown configuration by initially providing a mass of the material **40** to at least partially fill the opening **24** (i.e., to extend within the gaps **34**, and along outer edges **29** of the spacing structures **30**; with the mass being spaced from the outer edges by the dielectric barrier material **38**), and then removing some of the mass of material **40** while leave remaining portions of the material **40** within the gaps **34**.

The material **40** at the processing stage of FIG. **11** may be considered to be configured as segments **42** which are within the gaps **34**. Vertically-neighboring segments **42** are vertically-spaced from one another by intervening regions **44** which include the spacing structures **30**.

Referring to FIG. **12**, regions of the dielectric material **38** are removed from along the outer edges **29** of the spacing structures **30** to expose such outer edges along the opening **24**. The dielectric material **38** remaining at the processing stage of FIG. **12** may be considered to be configured as liners **46** extending within the cavities **34**, and along the peripheral surfaces **31**, **33** and **35** of such cavities.

Referring to FIG. **13**, the surfaces within opening **24** are exposed to oxidizing conditions which oxidize the segments **42** (FIG. **12**) to form charge-blocking regions **48**, and which also oxidize edges of the spacing structures **30** (FIG. **11**) to form the ledges **50**. The oxidizing conditions may utilize any suitable chemistry and operational parameters. In some example embodiments, the oxidizing conditions may comprise an operational temperature of the oxidizing ambient and/or the oxidizing surfaces of at least about 700° C. (but not limited to being at least about 700°, and may be lower if suitable oxidative conditions achieve desired electrical and/or other properties). The oxidizing conditions may, for example, utilize steam (for instance, in situ steam generation (ISSG)) as a source of the oxidant, and/or may utilize plasma to generate oxidizing species. The plasma may be a so-called "remote plasma", meaning that the plasma does not contact the surfaces within opening **24** which are to be oxidized, but instead only the oxidizing species generated by such plasma reach the oxidizing surfaces.

In some embodiments, the material **40** (FIG. **12**) of the segments **42** (FIG. **12**) comprises, consists essentially of, or consist of silicon nitride; and the material **32** (FIG. **12**) of the spacing structures **30** (FIG. **12**) comprises, consists essentially of, or consists of polycrystalline silicon. The oxidation may oxidize the silicon nitride **40** to form silicon oxynitride **54** and silicon dioxide **52**; and may oxidize the polycrystalline silicon **32** to form additional silicon dioxide **52**. In such embodiments, the charge-blocking regions **48** may comprise the silicon oxynitride **54** and the silicon dioxide **52** (as shown). Boundaries between the materials **52** and **54** are diagrammatically illustrated in FIG. **13** with dashed-lines **53** to indicate that the boundary between the materials **52** and **54** may be an abrupt interface between the silicon oxynitride and the silicon dioxide, or may be a gradient.

In some embodiments, at least portions of the charge-blocking regions **48** and/or the ledges **50** may be formed by deposition of suitable material(s) (e.g., silicon dioxide), followed by appropriate etching to achieve a desired shape (such as, for example, shapes analogous to the shapes of the charge-blocking regions **48** and ledges **50** shown in FIG. **13**).

The charge-blocking regions **48** extend vertically, and have horizontal thicknesses T1. Such horizontal thicknesses may be of any suitable dimension, and in some embodiments may be within a range of from about 50 Å to about 150 Å. The silicon oxynitride material **54** of the charge-blocking regions has a horizontal thickness T2; and the silicon dioxide material **52** of the charge-blocking regions has a horizontal thickness T3. In some embodiments, the horizontal thickness T2 will be at least about double the horizontal thickness T3. In some embodiments, the horizontal thickness T2 will be within a range of from about 20 Å to about 140 Å; and the horizontal thickness T3 will be within a range of from about 10 Å to about 30 Å. In some embodiments, the horizontal thicknesses T2 and T3 may be referred to as first and second horizontal thicknesses in order to distinguish them from one another.

The oxidation of materials **32** and **40** (FIG. **11**) may oxidize the polycrystalline silicon material **32** much faster than the silicon nitride material **40** (for instance, may oxidize the polycrystalline silicon at least 1.5 times as fast as the silicon nitride, at least twice as fast as the silicon nitride, at least three times as fast as the silicon nitride, etc.). In embodiments in which the oxidation induces much more expansion from the silicon dioxide **52** formed from the polycrystalline silicon **32** than from the silicon oxynitride **54**/silicon dioxide **52** formed from the silicon nitride material **40**, the edges of the tiers **16** along opening **24** may expand little, if at all, and the edges of the tiers **14** along opening **24** may expand substantially (e.g., the expansion due to the formation of the silicon dioxide **52** from polycrystalline silicon **32** material may be at least about double the expansion due to the formation of the silicon oxynitride **54**/silicon dioxide **52** from the silicon nitride material **40**). In some embodiments there may be substantial expansion along the edges of the tiers **14** along the opening **24**, and there may be substantially no expansion along the edges of the tiers **16** along the opening **24** (with the term "substantially no expansion" meaning no expansion to within reasonable tolerances of detection). The ledges **50** are shown to be horizontally longer than the charge-blocking regions **48**. Third gaps **56** are along the second levels **16**, and vertically between the ledges **50**.

In the shown embodiment, silicon oxynitride **58** is formed along regions where silicon dioxide **52** of ledges **50** is adjacent the silicon nitride **18** of levels **14**. Dashed-lines **59** are provided to show approximate boundaries between the silicon oxynitride **58** and the silicon nitride **18**, and to indicate that such boundaries may be abrupt interfaces, or may be gradients. In some embodiments, the ledges **50** may be considered to comprise front edges **55** along the opening **24**, and back edges **57** in opposing relation to the front edges **55**; with the back edges **57** being adjacent the first material **18**. The silicon oxynitride **58** is along the back edges **57** of the ledges **50**, and may be configured as silicon oxynitride liners **60**.

The silicon oxynitride **58** is optional, and in some embodiments may not be formed.

Referring to FIG. **14**, charge-storage material **62** is formed within the gaps **56**. The charge-storage material may comprise any suitable composition(s); and in some embodi-

ments may comprise charge-trapping materials, such as silicon nitride, silicon oxynitride, conductive nanodots, etc. Persons of ordinary skill in the art understand the term “charge-trapping”; and will understand that a “charge trap” may refer to an energy well that can reversibly capture a charge carrier (e.g., an electron or hole). In alternative embodiments (not shown), the charge-storage material may be configured as floating gate material (such as, for example, polycrystalline silicon).

The charge-storage material **62** may be formed in the shown configuration with any suitable methodology. For instance, in some embodiments the charge-storage material **62** may comprise, consist essentially of, or consist of silicon nitride, and may be initially formed to at least partially fill the opening **24**. Excess material **62** may then be removed with a suitable etch (for instance, an etch utilizing hot phosphoric acid, an etch utilizing oxidation followed by hydrofluoric acid treatment, etc.) to leave only the material **62** which is confined within the gaps **56**.

In some embodiments, the charge-storage material **62** may comprise silicon nitride, and the segments of the charge-storage material **62** remaining within the gaps **56** may be referred to as silicon nitride segments **64**.

Referring to FIG. **15**, edges of the silicon nitride segments **64** (FIG. **16**) are oxidized to form silicon oxynitride **66** and silicon dioxide **68**. Dashed lines **67** are utilized to illustrate approximate boundaries between the materials **66** and **68**. The material **62** remaining after the formation of the silicon oxynitride **66** and the silicon dioxide **68** may be considered to correspond to charge-storage regions (e.g., charge-trapping regions) **70**. The charge-trapping regions **70** have outer edges **71** adjacent the opening **24**, inner edges **73** adjacent the charge-blocking regions **48**, and horizontal edges **75** adjacent the ledges **50**. In the illustrated embodiment, the silicon oxynitride **66** extends along the outer edges **71** and along the horizontal edges **75**.

In some embodiments, the silicon oxynitride **66** may have a thickness within a range of from about 0.5 nm to about 3 nm, and the silicon dioxide **68** may have a thickness within a range of from about 0.5 nm to about 3 nm.

In the shown embodiment, the charge-storage regions **70** are along, and directly adjacent, the silicon dioxide **52** of the charge-blocking regions **48** (specifically, the inner edges **73** of the charge storage regions **70** are directly against such silicon dioxide **52**).

Referring to FIG. **16**, the ledges **50** (FIG. **15**) are removed to form fourth gaps **72**, with the fourth gaps being vertically between the charge-trapping regions **70**. In some embodiments, the material **52** (FIG. **15**) of the ledges **50** comprises silicon dioxide, and such material is removed selectively relative to other materials of construction **10** (for instance, by utilizing a buffered oxide etch).

Referring to FIG. **17**, material **74** is formed within the gaps **72**. In some embodiments, the material **74** may comprise, consist essentially of, or consist of silicon nitride; and may be considered to be configured as silicon nitride segments **76**. The silicon nitride segments **76** may be referred to as third silicon nitride segments. In some embodiments, the material **18** of the first levels **14** may be referred to as first silicon nitride, and the material **74** may be referred to as second silicon nitride which replaces some of the first silicon nitride.

The silicon nitride segments **74** may be formed with any suitable processing. For instance, opening **24** may be at least partially filled with silicon nitride, and then excess silicon nitride may be removed with appropriate etching to leave the segments **74** within the gaps **72**.

The silicon nitride segments **76** are along the first levels **14**, and are adjacent the first material **18** (which in some embodiments may be silicon nitride). In the shown embodiment, the silicon oxynitride **58** is between the silicon nitride segments **76** and the first material **18**. In other embodiments, the silicon oxynitride **58** may be omitted.

Referring to FIG. **18**, tunneling materials **78**, **80** and **82** are formed along an outer periphery of the opening **24**.

The tunneling materials can function as materials through which charge carriers tunnel or otherwise pass during programming operations, erasing operations, etc. In some contexts, one or more of the tunneling materials may be referred to as gate dielectric material, or simply as dielectric material. In the illustrated embodiment, three tunneling materials are utilized. In other embodiments, there may be fewer than three tunneling materials; and in yet other embodiments there may be more than three tunneling materials. In some embodiments, the tunneling materials **78**, **80** and **82** may be band-gap engineered to have desired charge tunneling properties. The tunneling material **80** is compositionally different from the materials **78** and **82**. The materials **78** and **82** may be compositionally different from one another in some embodiments, and may be compositionally the same as one another in other embodiments. In the illustrated embodiment, the silicon oxynitride **66** may correspond to an additional tunneling material. In some embodiments, one or more of the tunneling materials **78**, **80** and **82** may be omitted; and instead the function of such tunneling material may be encompassed with the silicon oxynitride **66**.

In some example embodiments, the tunneling material **80** may comprise silicon nitride, and the tunneling materials **78** and **82** may comprise silicon dioxide. In some example embodiments, the tunneling materials **66** and **78** may comprise silicon oxynitride and silicon dioxide, respectively; the tunneling material **80** may comprise silicon nitride, and the tunneling material **82** may comprise silicon dioxide.

In some embodiments, the tunneling materials **78**, **80** and **82** may be referred to as first, second and third tunneling materials, respectively.

Channel material **84** is formed within the opening **24** and along the tunneling materials **78**, **80** and **82**. In the illustrated embodiment, the channel material **84** is directly against the tunneling material **82**. The channel material **84** may comprise any suitable appropriately-doped semiconductor material(s); and in some embodiments may comprise one or more of silicon, germanium, III/V semiconductor materials (e.g., gallium phosphide), etc.

In the illustrated embodiment, the channel material **84** lines a periphery of the opening **24**, and insulative material **86** fills a remaining interior region of the opening **24**. The insulative material **86** may comprise any suitable composition or combination of compositions, such as, for example, silicon dioxide. The illustrated configuration of the channel material **84** may be considered to be a hollow channel configuration, in that the insulative material **86** is provided within a “hollow” in the channel configuration. In other embodiments, the channel material may be configured as a solid pillar.

The channel material **84** extends vertically along the periphery of opening **24**; or, in other words, extends vertically through the stack **12**.

Referring to FIG. **19**, second openings **88** are formed through the stack **12**. The second openings **88** extend through the portions of the first and second materials **18** and **20**.

FIG. **19A** shows a top view of a portion of the top level **16** of the illustrated region of construction **10**, and illustrates

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an example configuration in which the second openings **88** are configured as slits (i.e., trenches).

Referring to FIG. **20**, the material **20** (FIG. **19**) of the second levels **16** is removed to form cavities **90** along the second levels **16**. In some embodiments, the material **20** comprises silicon dioxide, and is removed with a buffered oxide etch.

Referring to FIG. **21**, conductive materials **92** and **94** are provided within the openings **88** and the cavities **90**. The conductive materials **92** and **94** may comprise any suitable electrically conductive composition(s), such as, for example, one or more of various metals (e.g., titanium, tungsten, cobalt, nickel, platinum, ruthenium, etc.), metal-containing compositions (e.g., metal silicide, metal nitride, metal carbide, etc.), and/or conductively-doped semiconductor materials (e.g., conductively-doped silicon, conductively-doped germanium, etc.). In some embodiments, the conductive material **92** may comprise, consist essentially of, or consist of titanium nitride; and the conductive material **94** may comprise, consist essentially of, or consist of tungsten.

Referring to FIG. **22**, conductive materials **92** and **94** are removed from within central regions of openings **88**, while leaving the conductive materials **92** and **94** within the cavities **90**.

The conductive materials **92** and **94** remaining within the cavities **90** together form conductive regions **96**. Although the illustrated conductive regions comprise two conductive materials, in other embodiments analogous conductive regions may comprise only a single conductive material, or may comprise more than two conductive materials.

Portions of the conductive regions **96** proximate the charge-blocking regions **48** may correspond to conductive gates **98**, and portions of the conductive regions **96** more distal from the charge-blocking regions **48** may correspond to wordlines **160**. The wordlines are along the levels **16**, and accordingly in some embodiments the levels **16** and may be referred to as wordline levels. Such wordline levels may be considered to alternate with the insulative levels **14** within the stack **12** of FIG. **22**.

In some embodiments, the conductive regions **96** may be considered to comprise first ends **161** and second ends **163** in opposing relation to the first ends. The first ends **161** are comprised by the gates **98**, and may be referred to as gate ends; and the second ends **163** are comprised by the wordlines **160**, and may be referred to as wordline ends.

Referring to FIG. **23**, the first material **18** (FIG. **22**), silicon oxynitride **58** (FIG. **22**), and material **74** (FIG. **22**) are removed with appropriate etching to leave voids **162**. In some embodiments, the materials **18** and **74** comprise, consist essentially of, or consist of silicon nitride, and the etching utilized to remove such materials comprises phosphoric acid as the primary etchant. The phosphoric acid will etch through the silicon oxynitride **58** slower than the silicon nitride materials; but can etch through the silicon oxynitride, particularly if the silicon oxynitride **58** is less than 10 Å thick. If the silicon oxynitride **58** is problematic, such may be omitted. Alternatively, the etching utilized to form the voids **162** may comprise multiple stages, with some of the stages being directed toward the removal of the silicon nitride materials **18** and **74**, and another stage being directed toward the removal of the silicon oxynitride material **58**.

A byproduct that may occur when utilizing phosphoric acid-based etches to remove silicon nitride is low-density silicon dioxide **164**. The term “low-density silicon dioxide” as utilized herein and in the claims that follow means silicon dioxide having an etch rate of at least 200 Å/minute when utilizing hydrofluoric acid diluted 1:50 in water, at room

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temperature. In some embodiments, the low-density silicon dioxide may have an etch rate of at least about 300 Å/minute, at least about 500 Å/minute, or even at least about 1000 Å/minute; when utilizing hydrofluoric acid diluted 1:50 in water, at room temperature. Low-density silicon dioxide is highly porous, and accordingly has a lower dielectric constant than higher density (less porous) silicon dioxide. It can be desired to utilize low dielectric constant materials as insulative materials between vertically-adjacent wordlines in order to reduce capacitive coupling between the wordlines.

Some embodiments advantageously utilized the low-density silicon dioxide **164** to seal the voids **162**, and such is diagrammatically illustrated in FIG. **23**. The low-density silicon dioxide **164** produced as a byproduct of the etching of the silicon nitride wraps around the wordline ends **163** of the conductive regions **96**. The low-density silicon dioxide **164** may be considered to be configured as insulative liner regions **166** which extend around the ends **163** of the conductive regions **96**. In the embodiment of FIG. **23**, the insulative liner regions **166** merge with one another to pinch off ends of the voids **162** (or, in other words, join with one another to seal ends of the voids).

The insulative liners **166** do not entirely fill the openings **88** in the embodiment of FIG. **23**. Accordingly, remaining portions of the openings **88** are filled with additional insulative material **168**, as shown in FIG. **24**. The insulative material **168** may comprise any suitable composition(s). In some embodiments, the insulative material **168** will comprise, consist essentially of, or consist of silicon dioxide; but will be of a higher density than the low-density silicon dioxide **164**. For instance, in some embodiments the insulative material **168** may have an etch rate of less than about 100 Å/minute when utilizing hydrofluoric acid diluted 1:50 in water, at room temperature.

The embodiment of FIG. **23** shows low-density insulative material **164** forming insulative regions **166** which merge with one another to seal the voids **162**. In other embodiments, at least some of the insulative regions may not merge with one another; and accordingly, open orifices **170** may remain at ends of the voids (i.e., may extend into the voids), as shown in FIG. **25**. However, the insulative material **168** utilized to fill the openings **88** may also seal the orifices **170**, as shown in FIG. **26**.

The assemblies **10** of FIGS. **24** and **26** comprise example arrangements of memory cells, and in some embodiments may correspond to example configurations of memory arrays (e.g., NAND memory arrays). Specifically, the conductive gates **98**, together with the dielectric barrier material **38**, charge-blocking regions **48**, charge-storage regions **70**, tunneling materials **66**, **78**, **80** and **82**, and channel material **84** may be incorporated into memory cells **180a**, **180b** and **180c**. Such memory cells may be NAND memory cells in some embodiments. The illustrated memory cells are vertically-stacked one atop another, and may be part of a NAND string. The memory cells **180a**, **180b** and **180c** are substantially identical to one another (with the term “substantially identical” meaning identical to within reasonable tolerances of fabrication and measurement), and in some embodiments may be referred to as a first memory cell, a second memory cell and a third memory cell, respectively. The memory cells **180a**, **180b** and **180c** may be considered to be representative of a large number of substantially identical memory cells which may be fabricated across a memory array; such as, for example, a NAND memory array analogous to those described above with reference to FIGS. **1-4**.

In operation, the charge-storage regions **70** may be configured to store information in the memory cells **180a**, **180b** and **180c**. The value (with the term “value” representing one bit or multiple bits) of information stored in an individual memory cell (e.g., **180a**) may be based on the amount of charge (e.g., the number of electrons) stored in a charge-storage region. The amount of charge within an individual charge-storage region **70** may be controlled (e.g., increased or decreased), at least in part, based on the value of voltage applied to an associated gate **98**, and/or based on the value of voltage applied to an associated channel material **84**.

The tunneling materials **66**, **78**, **80** and **82** together form tunneling regions **184** of the memory cells **180a**, **180b** and **180c**. Such tunneling regions may be configured to allow desired tunneling (e.g., transportation) of charge (e.g., electrons) between the charge-storage regions **70** and the channel material **84**. The tunneling regions **184** may be configured (i.e., engineered) to achieve a selected criterion, such as, for example, but not limited to, an equivalent oxide thickness (EOT). The EOT quantifies the electrical properties of the tunneling region (e.g., capacitance) in terms of a representative physical thickness. For example, EOT may be defined as the thickness of a theoretical silicon dioxide layer that would be required to have the same capacitance density as a given dielectric (e.g., tunneling region **184**), ignoring leakage current and reliability considerations.

The charge-blocking regions **48** are adjacent to the charge-storage regions **70**, and may provide a mechanism to block charge from flowing from the charge-storage regions **70** to the associated gates **98**. The dielectric barrier material **38** is provided between the charge-blocking regions **48** and the associated gates **98**, and may be utilized to inhibit back-tunneling of electrons from the gates **98** toward the charge-storage regions **70**. In some embodiments, the dielectric barrier material **38** may be considered to form dielectric barrier regions within the memory cells **180a**, **180b** and **180c**.

In some embodiments, the channel material **84** may be considered to be a channel configured to conduct current. The channel includes a first channel portion **190** which is incorporated into the first memory cell **180a**, a second channel portion **192** which is incorporated into the second memory cell **180b**, and a third channel portion **194** which is incorporated into the third memory cell **180c**.

The gate **98** which is incorporated into the memory cell **180a** may be referred to as a first gate, the gate **98** which is incorporated into the second memory cell **180b** may be referred to as a second gate, and the gate **98** which is incorporated into the third memory cell **180c** may be referred to as a third gate.

The charge-blocking region **48** and charge-storage region **70** within a memory cell may be together considered to be a memory cell structure **196**. The memory cell structure **196** within the first memory cell **180a** may be referred to as a first memory cell structure, the memory cell structure **196** within the second memory cell **180b** may be referred to as a second memory cell structure, and the memory cell structure **196** within the third memory cell **180c** may be referred to as a third memory cell structure. The dielectric barrier material **38** forms dielectric barrier regions within the first, second and third memory cells **180a**, **180b** and **180c**. Such dielectric barrier regions may be considered to be part of the memory cell structures **196** in some embodiments, and may be considered to be separate from the memory cell structures **196** in other embodiments. The dielectric barrier region within the first memory cell **180a** may be referred to as a first dielectric barrier region, the dielectric barrier region within

the second memory cell **180b** may be referred to as a second dielectric barrier region, and the dielectric barrier region within the third memory cell **180c** may be referred to as a third dielectric barrier region.

The charge-blocking regions **48** within the memory cell structures **196** are between the charge-storage regions **70** and the gates **98**. In the illustrated embodiments of FIGS. **24** and **26**, the silicon oxynitride **54** within the charge-blocking regions **48** may be considered to have two opposing lateral sides **193** and **195**. The silicon dioxide **52** of the charge-blocking regions **48** is along one of the opposing lateral sides (the side **193**), and the other of the opposing lateral sides (the side **195**) is directly adjacent the dielectric barrier material **38** in the shown embodiment.

The gate **98** of the first memory cell **180a** is vertically spaced from the gate **98** of the second memory cell **180b**. One of the voids **162** is vertically between the vertically-spaced gates **98** of the first and second memory cells **180a** and **180b**; and such void is labeled as **162a** in FIGS. **24** and **26** to enable the void to be distinguished from the other voids.

The void **162a** has a region **187** between the vertically-neighboring gates of the memory cells **180a** and **180b**, and has another region **189** between the vertically-neighboring memory cell structures **196** of the memory cells **180a** and **180b**. The other voids **162** have similar regions **187** and **189**. Accordingly, the voids **162** can provide electrical isolation between vertically-neighboring memory cell structures **196**, and can also provide electrical isolation between vertically-neighboring conductive gates **98**. In the shown embodiment of FIGS. **24** and **26**, at least portions of the conductive regions **96** of the gates **98** are spaced from the voids **162** by low-density silicon dioxide **164**, and the charge-storage regions **70** are spaced from the voids by liners **199** corresponding to regions of the silicon oxynitride **66**. The dielectric barrier materials **38** are directly against the voids **162**.

In some embodiments, the stacks **12** of FIGS. **24** and **26** may be considered to comprise alternating insulative levels **14** and wordline levels **16**. The channel material **84** extends vertically along the stack **12**, and the gates **98** are along the wordline levels and spaced from the channel material by the tunneling materials (**66**, **76**, **80** and **82**), the memory cell structures **196**, and the dielectric barrier material **38**.

The charge-storage regions (i.e., charge-storage structures) **70** are vertically-spaced from one another by intervening regions of the voids **162** along the insulative levels **14**. The vertical separation of the charge-storage regions **70** from one another may alleviate or prevent charge leakage between neighboring charge-storage regions within a common NAND string; and may mitigate coupling of charge-storage regions with other components (e.g., neighboring charge-storage regions, control gate, channel, tunnel oxide, etc.). Such may enable substantial improvements relative to conventional NAND configurations having a continuous charge-storage structure extending along all of the memory cells of a NAND string. Example improvements may include one or more of improved endurance, improvement in read/write budget, improvement in quick charge gain, improvement in quick charge loss, reduced cell-to-cell capacitive coupling, cycling, etc.

Isolation of vertically-adjacent charge-storage regions from one another (as compared to conventional configurations having continuous charge-storage regions) may alleviate or prevent charge leakage and charge coupling between neighboring charge-storage regions, and other neighboring components within a common NAND string.

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Provision of voids (e.g., air gaps) between vertically-adjacent wordlines may reduce capacitive capacitance between the wordlines, and may improve resistor-capacitor (RC) properties of the wordlines. Such may reduce programming delay and/or provide other advantages as compared to other configurations.

The assemblies and structures discussed above may be utilized within integrated circuits (with the term “integrated circuit” meaning an electronic circuit supported by a semiconductor substrate); and may be incorporated into electronic systems. Such electronic systems may be used in, for example, memory modules, device drivers, power modules, communication modems, processor modules, and application-specific modules, and may include multilayer, multi-chip modules. The electronic systems may be any of a broad range of systems, such as, for example, cameras, wireless devices, displays, chip sets, set top boxes, games, lighting, vehicles, clocks, televisions, cell phones, personal computers, automobiles, industrial control systems, aircraft, etc.

Unless specified otherwise, the various materials, substances, compositions, etc. described herein may be formed with any suitable methodologies, either now known or yet to be developed, including, for example, atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), etc.

The terms “dielectric” and “insulative” may be utilized to describe materials having insulative electrical properties. The terms are considered synonymous in this disclosure. The utilization of the term “dielectric” in some instances, and the term “insulative” (or “electrically insulative”) in other instances, may be to provide language variation within this disclosure to simplify antecedent basis within the claims that follow, and is not utilized to indicate any significant chemical or electrical differences.

The particular orientation of the various embodiments in the drawings is for illustrative purposes only, and the embodiments may be rotated relative to the shown orientations in some applications. The descriptions provided herein, and the claims that follow, pertain to any structures that have the described relationships between various features, regardless of whether the structures are in the particular orientation of the drawings, or are rotated relative to such orientation.

The cross-sectional views of the accompanying illustrations only show features within the planes of the cross-sections, and do not show materials behind the planes of the cross-sections, unless indicated otherwise, in order to simplify the drawings.

When a structure is referred to above as being “on”, “adjacent” or “against” another structure, it can be directly on the other structure or intervening structures may also be present. In contrast, when a structure is referred to as being “directly on”, “directly adjacent” or “directly against” another structure, there are no intervening structures present.

Structures (e.g., layers, materials, etc.) may be referred to as “extending vertically” to indicate that the structures generally extend upwardly from an underlying base (e.g., substrate). The vertically-extending structures may extend substantially orthogonally relative to an upper surface of the base, or not.

Some embodiments include a memory array which has a vertical stack of alternating insulative levels and wordline levels. Channel material extends vertically along the stack. Conductive segments are along the wordline levels. Individual of the conductive segments have, along a cross-section, first and second ends in opposing relation to one another. The conductive segments include gates and wordlines adjacent the gates. The wordlines encompass the

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second ends and the gates encompass the first ends. Memory cell structures are along the wordline levels and are located between the gates and the channel material. The memory cell structures include charge-storage regions and charge-blocking regions. The charge-blocking regions are between the charge-storage regions and the gates. Voids are along the insulative levels and are between vertically-neighboring of the memory cell structures. Insulative liner regions are along the wordline levels and extend around the second ends of the conductive segments. The insulative liner regions contain low density silicon dioxide.

Some embodiments include an assembly having a channel to conduct current. The channel includes a first channel portion and a second channel portion under the first channel portion. A first memory cell structure is located between a first gate and the first channel portion. The first memory cell structure includes a first charge-storage region and a first charge-blocking region. The first charge-blocking region is located between the first charge-storage region and the first gate. The first charge-blocking region comprises silicon oxynitride, and the first charge-storage region comprises silicon nitride. A second memory cell structure is under the first memory cell structure and is located between a second gate and the second channel portion. The second memory cell structure includes a second charge-storage region and a second charge-blocking region. The second charge-blocking region is located between the second charge-storage region and the second gate. The second charge-blocking region comprises silicon oxynitride, and the second charge-storage region comprises silicon nitride. A void is located between the first and second gates, and between the first and second memory cell structures. The first gate comprises conductive material over the void and spaced from the void by low-density silicon dioxide. The second gate comprises conductive material under the void and spaced from the void by low-density silicon dioxide. A first liner is between the silicon nitride of the first charge-storage region and the void. A second liner is between the silicon nitride of the first charge-storage region and the void.

Some embodiments include a method of forming an assembly. A first opening is formed through a stack of alternating first and second levels. The first levels contain silicon nitride, and the second levels contain silicon dioxide. Some of the silicon dioxide of the second levels is replaced with memory cell structures while leaving remaining portions of the silicon dioxide of the second levels. The memory cell structures include charge-storage regions adjacent charge-blocking regions. Vertically-extending tunneling material is formed within the first opening. The tunneling material extends along the charge-storage regions of the memory cell structures. Channel material is formed within the first opening and adjacent the tunneling material. A second opening is formed through the stack. The second opening extends through the remaining portions of the silicon dioxide of the second levels, and extends through the silicon nitride of the first levels. The remaining portions of the silicon dioxide of the second levels are removed to form cavities along the second opening. Conductive regions are formed within the cavities. The silicon nitride of the first levels is removed to form voids between the conductive regions.

Some embodiments include a method of forming an assembly. A first opening is formed through a stack of alternating first and second levels. The first levels comprise first material, and the second levels comprise second material. The first material of the first levels is recessed along the first opening to form first gaps. The first gaps are vertically

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between segments of the second levels. Spacing structures are formed within the first gaps. The second material of the second levels is recessed along the first opening to form second gaps and to leave remaining portions of the second material along the second gaps. The second gaps are vertically between segments of the spacing structures. Individual of the second gaps each have an upper peripheral surface, a lower peripheral surface and an inner peripheral surface. Liners of dielectric barrier material are formed to extend along the upper peripheral surface, the inner peripheral surface and the lower peripheral surface of each of the second gaps, and to narrow the second gaps. First silicon nitride segments are formed within the narrowed second gaps and along the second levels. Vertically-neighboring of the first silicon nitride segments are vertically-spaced from one another by intervening regions which include the spacing structures. Oxidizing conditions are utilized to oxidize the silicon nitride segments and to oxidize the spacing structures. The oxidized silicon nitride segments are charge-blocking regions. The oxidized spacing structures are ledges. Third gaps are vertically between the ledges. Second silicon nitride segments are formed within the third gaps and along the charge-blocking regions. Edges of the second silicon nitride segments are oxidized. The oxidized edges include edges along the first opening, and include edges along the ledges. Remaining regions of the silicon nitride segments which are not oxidized are charge-trapping regions. The ledges are removed to leave fourth gaps. The fourth gaps are vertically between the charge-trapping regions. Third silicon nitride segments are formed within the fourth gaps. The third silicon nitride segments are along the first levels and are adjacent remaining portions of the first material. Vertically-extending tunneling material is formed within the first opening. The tunneling material extends along edges of the third silicon nitride segments and along the charge-trapping regions. Channel material is formed within the first opening and adjacent the tunneling material. A second opening is formed through the stack. The second opening extends through the remaining portions of the first material, and through the remaining portions of the second material. The remaining portions of the second material of the second levels are removed to form cavities along the second levels. Conductive regions are formed within the cavities. The remaining portions of the first material and the third silicon nitride segments of the first levels are removed to form voids along the second opening.

In compliance with the statute, the subject matter disclosed herein has been described in language more or less specific as to structural and methodical features. It is to be understood, however, that the claims are not limited to the specific features shown and described, since the means herein disclosed comprise example embodiments. The claims are thus to be afforded full scope as literally worded, and to be appropriately interpreted in accordance with the doctrine of equivalents.

We claim:

1. An assembly, comprising:

a channel region including a first channel portion and a second channel portion under the first channel portion; a tunneling material extending along an entirety of the channel region;

a first memory cell structure located between a first gate and the first channel portion; the first memory cell structure including a first charge-storage region and a first charge-blocking region; the first charge-blocking region being located between the first charge-storage region and the first gate;

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a second memory cell structure under the first memory cell structure and located between a second gate and the second channel portion; the second memory cell structure including a second charge-storage region and a second charge-blocking region; the second charge-blocking region being located between the second charge-storage region and the second gate;

a void between the first and second gates, and between the first and second memory cell structures and extending to the tunneling material;

a first liner between the silicon nitride of the first charge-storage region and the void;

a second liner between the silicon nitride of the second charge-storage region and the void;

a first dielectric barrier region being between the first gate and the first charge-blocking region, the first dielectric barrier region being directly against the void along a lower surface of the first charge-blocking region;

a second dielectric barrier region being between the second gate and the second charge-blocking region, the second dielectric barrier region being directly against the void along an upper surface of the second charge-blocking region;

a first region of a low-density silicon dioxide between the first gate and the void;

a second region of the low-density silicon dioxide between the second gate and the void;

an insulative material along and directly contacting a portion of the first region of low-density silicon dioxide and along and directly contacting a portion of the second regions of low-density silicon dioxide, the insulative material comprising silicon dioxide having a higher density than the low-density silicon dioxide of the first and second regions; and

wherein an edge of the first dielectric barrier region is directly against the void; and wherein an edge of the second dielectric barrier region is directly against the void, and wherein the first region of low-density silicon dioxide does not contact the second region of low-density silicon dioxide.

2. The assembly of claim 1 wherein the first and second liners comprise silicon oxynitride.

3. The assembly of claim 1 wherein the first and second charge-blocking regions comprise silicon oxynitride and the first and second charge-storage regions comprise silicon nitride.

4. The assembly of claim 3 wherein the first charge-blocking region comprises silicon dioxide between the silicon oxynitride of the first charge-blocking region and the silicon nitride of the first charge-storage region; and wherein the second charge-blocking region comprises silicon dioxide between the silicon oxynitride of the second charge-blocking region and the silicon nitride of the second charge-storage region.

5. A method of forming an assembly, comprising:

forming a channel region having a first channel portion and a second channel portion under the first channel portion;

forming a first memory cell structure located between a first gate and the first channel portion; the first memory cell structure including a first charge-storage region and a first charge-blocking region; the first charge-blocking region being located between the first charge-storage region and the first gate;

forming a second memory cell structure under the first memory cell structure and located between a second gate and the second channel portion;

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the second memory cell structure including a second charge-storage region and a second charge-blocking region; the second charge-blocking region being located between the second charge-storage region and the second gate;

forming a void between the first and second gates, and between the first and second memory cell structures;

forming a first liner between the silicon nitride of the first charge-storage region and the void;

forming a second liner between the silicon nitride of the second charge-storage region and the void;

forming a high-k dielectric barrier material between the first gate and the first charge-blocking region, the first high-k dielectric barrier material being directly against the void along a lower surface of the first charge-blocking region;

forming the high-k dielectric barrier material between the second gate and the second charge-blocking region, the high-k dielectric barrier material being directly against the void along an upper surface of the second charge-blocking region;

forming a first region of a low-density silicon dioxide between the first gate and the void;

forming a second region of the low-density silicon dioxide between the second gate and the void; wherein the

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first region of low-density silicon dioxide does not contact the second region of low-density silicon dioxide; and

forming an insulative material along and directly contacting a portion of the first region of low-density silicon dioxide and along and directly contacting a portion of the second regions of low-density silicon dioxide, the insulative material comprising silicon dioxide having a higher density than the low-density silicon dioxide of the first and second regions.

6. The method of claim 5 wherein the first and second liners comprise silicon oxynitride.

7. The method of claim 5 wherein the first and second charge-blocking regions comprise silicon oxynitride and the first and second charge-storage regions comprise silicon nitride.

8. The method of claim 7 wherein the first charge-blocking region comprises silicon dioxide between the silicon oxynitride of the first charge-blocking region and the silicon nitride of the first charge-storage region; and wherein the second charge-blocking region comprises silicon dioxide between the silicon oxynitride of the second charge-blocking region and the silicon nitride of the second charge-storage region.

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